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## Chat history

**You said:**

die hard test list for randomness

**ChatGPT said:**

Here's the **Diehard (Marsaglia) test suite** list (commonly cited as **15 tests**) used for **randomness testing**:

1. **Birthday Spacings**
2. **Overlapping 5-Permutations**
3. **Binary Rank Test (31×31 matrices)**
4. **Binary Rank Test (32×32 matrices)**
5. **Binary Rank Test (6×8 matrices)**
6. **Bitstream Test**
7. **Overlapping Pairs Sparse Occupancy (OPSO)**
8. **Overlapping Quadruples Sparse Occupancy (OQSO)**
9. **DNA Test**
10. **Count-the-1's Test (on a stream of bytes)**
11. **Count-the-1's Test (on specific bytes)**
12. **Parking Lot Test**
13. **Minimum Distance Test**
14. **3D Spheres Test**
15. **Squeeze Test**

If you want, I can also list the **Dieharder** extended suite (many more tests) or tell you **what input format** (bits/32-bit ints) each Diehard test expects.

**You said:**

tell the easiest test first

**ChatGPT said:**

If you want the **easiest Diehard tests to implement first**, start with the ones that are basically **counting + simple distributions** (no big matrices / geometry / heavy memory):

**Easiest → Harder (good beginner order)**

1. **Count-the-1's (stream of bytes)**
  - Just count how many 1-bits appear in bytes/blocks and compare to what you expect.

## 2. Count-the-1's (specific bytes)

- Same idea, just grouped differently.

## 3. Squeeze test

- Repeatedly multiply uniforms until you “squeeze” past 1; compare the number of steps to the expected distribution.

## 4. Birthday spacings

- Generate many integers, sort, take spacings, count repeats (Poisson approx).

## 5. Minimum distance test

- Put random points in 2D, compute nearest-neighbor distances; compare to expected.

## 6. Parking lot test (*conceptually easy, implementation a bit longer*)

- Randomly “park” non-overlapping circles; count how many fit.

Usually *not* the first ones to code

- **Binary rank tests (31×31, 32×32, 6×8)** (need bit-matrix rank over GF(2))
- **Bitstream / OPSO / OQSO / DNA** (more specialized + careful indexing)
- **3D spheres** (geometry + performance)

If you tell me what your RNG outputs (bits, bytes, 32-bit ints, or doubles), I'll suggest the best “first 3” and the exact input parsing for them.

**You said:**

and tell how much bytes or bits it takes each test

**ChatGPT said:**

Thought for 2m 18s

Below is the **data size each Diehard test consumes** in the *standard Marsaglia-style parameter settings*, assuming your input is a file of **32-bit unsigned integers** (so **1 integer = 32 bits = 4 bytes**). Where a test restarts the file for many bit-windows, I show both **per window** and the **full “all windows” total**.

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**Diehard tests: how many bits/bytes they use**

### 1) Birthday spacings

- Uses  $m = 2^9 = 512$  “birthdays”, repeated **500** times → **256,000 integers per p-value**
- Does **9 bit-windows** (bits 1–24, 2–25, ..., 9–32)
- **Total:**  $256,000 \times 9 = 2,304,000$  ints = **9,216,000 bytes (~8.8 MiB)**

### 2) Overlapping 5-permutations (OPERM5)

- “Uses **1,000,000 integers, twice**”

- **Total:** 2,000,000 ints = **8,000,000 bytes (~7.6 MiB)**

### 3) Binary rank 31×31

- **40,000 matrices**, each needs **31 integers**
- **Total:**  $40,000 \times 31 = 1,240,000$  ints = **4,960,000 bytes (~4.7 MiB)**

### 4) Binary rank 32×32

- **40,000 matrices**, each needs **32 integers**
- **Total:** 1,280,000 ints = **5,120,000 bytes (~4.9 MiB)**

### 5) Binary rank 6×8

- **100,000 matrices**, each uses **6 integers** (one chosen byte from each)
- **Total:** 600,000 ints = **2,400,000 bytes (~2.3 MiB)**

### 6) Bitstream

- Needs a bitstring of length  $2^{21} + 19$  bits per run
- Repeated **20 times**
- **Total bits:**  $20 \times (2,097,171) = 41,943,420$  bits = **5,242,927.5 bytes (~5.0 MiB)**

### 7) OPSO (Overlapping Pairs Sparse Occupancy)

- Builds  $2^{21}$  overlapping 2-letter words from  $2^{21} + 1$  “keystrokes”
- Each keystroke comes from **one 32-bit integer** (taking 10 chosen bits)
- **Per 10-bit window:**  $(2^{21}+1)$  ints  $\approx 2,097,153$  ints  $\approx 8,388,612$  bytes (**~8.0 MiB**)
- Restarts for **all 10-bit windows:**  $32-10+1 = 23$  windows
- **All windows total:**  $\sim 23 \times 8.39 \text{ MB} \approx 193 \text{ MB}$

### 8) OQSO (Overlapping Quadruples Sparse Occupancy)

- Uses  $2^{21}$  4-letter words from  $2^{21} + 3$  keystrokes
- Each letter is from **5 consecutive bits** of one 32-bit integer
- **Per 5-bit window:**  $(2^{21}+3)$  ints  $\approx 2,097,155$  ints  $\approx 8,388,620$  bytes (**~8.0 MiB**)
- All 5-bit windows:  $32-5+1 = 28$  windows
- **All windows total:**  $\sim 28 \times 8.39 \text{ MB} \approx 235 \text{ MB}$

### 9) DNA

- Uses  $2^{21}$  overlapping 10-letter words from  $2^{21} + 9$  keystrokes
- Each “letter” is from **2 bits**

- **Per 2-bit window:**  $(2^{21}+9)$  ints  $\approx 2,097,161$  ints  $\approx 8,388,644$  bytes ( $\sim 8.0$  MiB)
  - If you do all consecutive 2-bit windows:  $32-2+1 = 31$  windows
  - **All windows total:**  $\sim 31 \times 8.39$  MB  $\approx 260$  MB
- 

#### 10) Count-the-1's (stream of bytes)

- Uses **256,000 overlapping 5-letter words** (letters are bytes)
- Needs  $256,000 + 4 = 256,004$  bytes  $\approx 0.244$  MiB

#### 11) Count-the-1's (chosen bytes)

- Same **256,000** overlapping words, but each letter comes from a chosen byte of **one 32-bit integer**
  - Needs **256,004 integers** = **1,024,016 bytes** ( $\sim 1.0$  MiB)
- 

#### 12) Parking lot

- **12,000** tries per run
- Typically needs **2 uniforms per try** (x,y)  $\Rightarrow$  about **24,000 ints per run**  $\approx 96,000$  bytes
- KS sample size is **10 runs**
- **Total:**  $\sim 240,000$  ints = **960,000 bytes** ( $\sim 0.92$  MiB)

#### 13) Minimum distance

- Done **100 times**, each time uses **8000 points** in 2D
- 2D point needs **2 uniforms**  $\Rightarrow \sim 16,000$  ints per trial
- **Total:**  $100 \times 16,000 = 1,600,000$  ints = **6,400,000 bytes** ( $\sim 6.1$  MiB)

#### 14) 3D spheres

- Uses **4000 points**, repeated **20 times**
- 3D point needs **3 uniforms**  $\Rightarrow 12,000$  ints per repeat
- **Total:**  $20 \times 12,000 = 240,000$  ints = **960,000 bytes** ( $\sim 0.92$  MiB)

#### 15) Squeeze

- Produces **100,000 "j"** values (each j uses a variable number of uniforms)
- So input size is **variable**:

$$\text{\text{\#ints}} = \sum_{i=1}^{100000} j_i$$

(often ends up being **a few million 32-bit ints**, i.e., **tens of MB**, depending on the run)

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If you tell me your file format (**raw uint32 binary**, or ASCII numbers, or bytes), I can convert everything into an exact “minimum file size you must generate” for your setup.

**You said:**

explain birthday test in easiest with example

**ChatGPT said:**

Sure — **Diehard “Birthday Spacings”** is basically: *generate many random “birthdays”, sort them, look at the gaps between them, and count how often the same gap repeats*. Too many repeated gaps  $\Rightarrow$  not random.

**The idea (in plain words)**

- Imagine a year with **N** possible days (like  $1 \dots N$ ).
- Pick **m** random days (these are the “birthdays”).
- Sort them.
- Compute the **spacings** (differences) between consecutive sorted days.
- Now count **how many spacings are repeated** (same gap value appearing more than once).
- For a good RNG, the number of repeated spacings behaves like a **Poisson** random variable with a predictable mean.

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**Tiny example (easy to see)**

Let’s take a toy range **N = 50** and pick **m = 8** birthdays:

Random birthdays (unsorted):

[3, 11, 12, 20, 27, 29, 41, 46]

Sorted (already sorted here):

3, 11, 12, 20, 27, 29, 41, 46

Compute spacings (differences):

- $11 - 3 = 8$
- $12 - 11 = 1$
- $20 - 12 = 8$
- $27 - 20 = 7$
- $29 - 27 = 2$
- $41 - 29 = 12$
- $46 - 41 = 5$

Spacings list:

[8, 1, 8, 7, 2, 12, 5]

Now look for repeats:

- spacing **8** occurs **twice** → that's a repeated spacing.

So here the "collision count" is:

- $K = 1$  repeated spacing value (or sometimes counted as "number of repeats", same idea)

If your RNG is biased/structured, you'll tend to see **too many repeated spacings** across many trials.

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### What Diehard actually does (standard settings)

Diehard uses large parameters so statistics become sharp:

- Choose  $m = 512$  birthdays per trial
- Range is  $N = 2^{24}$  (i.e., birthdays are 24-bit integers)
- Repeat this trial **500 times**
- For each trial, compute spacings and count repeats → sum over 500 trials to get  $K$

### Why Poisson?

For truly random picks, the expected number of repeated spacings is approximately:

$$\lambda \approx \frac{m^3}{4N}$$

For Diehard's values  $m = 512$ ,  $N = 2^{24}$ :

- $m^3 = 512^3 = 134,217,728$
  - $4N = 4 \cdot 2^{24} = 67,108,864$
- So

$$\lambda \approx \frac{134,217,728}{67,108,864} \approx 2$$

That means: **the number of repeated spacings per trial behaves like Poisson( $\approx 2$ )**.

Over **500 trials**, the total behaves like Poisson( $\approx 1000$ ).

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### How you compute the p-value

Let:

- observed total repeats =  $K_{\text{obs}}$
- expected mean =  $\Lambda = 500\lambda \approx 1000$

Then:

$$p = \Pr(\text{Poisson}(\Lambda) \geq K_{\text{obs}})$$

(or you can do a 2-sided version, checking if it's *too small or too large*).

If **p** is **extremely small** (like  $< 1e-6$ ) or extremely close to 1, that's suspicious.

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### Implementation steps (exactly what to code)

For each trial (repeat 500 times):

1. Generate  $m=512$  random 24-bit integers in  $[0, 2^{24})$ .
2. Sort them.
3. Compute  $sp[i] = x[i+1] - x[i]$  for  $i=0..m-2$ .
4. Sort the spacings.
5. Count duplicates in the sorted spacings:
  - if  $sp[i] == sp[i-1]$ , increment repeat counter.
6. Add repeats to total K.

That's it.

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If you want, I can give you a **clean C function** that:

- reads uint32 from your .dat file,
- extracts 24-bit values (or does the "sliding window" version Diehard uses),
- outputs  $K$ ,  $\Lambda$ , and the p-value.

**You said:**

not get it please

**ChatGPT said:**

No worries — I'll explain it like you're doing it with pen + paper.

### What the Birthday Spacings test is checking

It checks **whether your random numbers create "too many repeated gaps"** after sorting.

Random numbers should look "messy".

If they have patterns, you start seeing the **same gap size** again and again more than expected.

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### Step-by-step with a super small example

#### Step 1: pick random "birthdays"

Pretend a "year" has only **20 days** (0 to 19).

Pick **8 random birthdays**:

[2, 15, 7, 8, 11, 4, 18, 6]

### Step 2: sort them

Sorted:

[2, 4, 6, 7, 8, 11, 15, 18]

### Step 3: compute the gaps (spacings)

Subtract neighbors:

- $4 - 2 = 2$
- $6 - 4 = 2$
- $7 - 6 = 1$
- $8 - 7 = 1$
- $11 - 8 = 3$
- $15 - 11 = 4$
- $18 - 15 = 3$

So spacings are:

[2, 2, 1, 1, 3, 4, 3]

### Step 4: count repeated spacings

Look for duplicates:

- spacing **2** appears twice → repeat
- spacing **1** appears twice → repeat
- spacing **3** appears twice → repeat

So repeats = **3** (there are 3 duplicated gap values)

✓ That's the whole "birthday spacings" idea.

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### Why is this a randomness test?

If numbers are truly random, after sorting, the gaps should vary naturally.

You'll get **some** repeated gaps, but not *too many*.

If your RNG has structure (like numbers falling into patterns), you get **lots of equal gaps**, which is suspicious.

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### How Diehard uses it (same steps, just bigger)

Diehard does exactly the same thing, but:

- takes **m = 512** random numbers



- each number is a “birthday” in a huge range (0 to  $2^{24} - 1$ )
- repeats this **500 times**
- adds up all repeats into one big count  $K$
- compares  $K$  to what a good RNG should produce

If  $K$  is way too high or way too low → RNG fails.

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### One-line summary

**Birthday Spacings = sort random numbers → take neighbor differences → count how often the same difference repeats.**

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If you tell me what you generate (bytes / uint32 / bits), I’ll write the exact tiny C code for this test in the simplest way (no “bit windows”, just the core test).

**You said:**

now i dsidint get how diehard will work on my byte file

**ChatGPT said:**

Got it — you have a **raw byte file** (like rng.dat) and you’re confused how Diehard reads it.

**Key point**

**Classic Diehard expects a stream of 32-bit integers.**

So your **byte file is consumed 4 bytes at a time** → that becomes **one uint32** number.

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### 1) How Diehard reads your byte file

If your file is:

b0 b1 b2 b3 b4 b5 b6 b7 ...

Diehard groups it like:

- $\text{Int0} = (\text{b0}, \text{b1}, \text{b2}, \text{b3})$
- $\text{Int1} = (\text{b4}, \text{b5}, \text{b6}, \text{b7})$
- $\text{Int2} = (\text{b8}, \text{b9}, \text{b10}, \text{b11})$
- ...

So:

- **4 bytes = 1 number = 32 bits**
- If you have **X bytes**, you have  **$X/4$  uint32 numbers**

**Endianness note (important):**

Most PC implementations treat those 4 bytes as **little-endian**:

$$x = b_0 + 256b_1 + 256^2b_2 + 256^3b_3$$

(If you reverse byte order, your numbers change, and the test results can change.)

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**2) Now: how Birthday Spacings uses those numbers**

Birthday Spacings needs “birthdays” in a fixed range, usually:

- **$N = 2^{24}$**  (so each birthday is a **24-bit integer**)

But your input numbers are 32-bit. So Diehard **extracts 24 bits** from each 32-bit int.

**The simplest extraction (one window)**

Take the **lowest 24 bits**:

$$b = x \& (2^{24} - 1)$$

That gives a birthday  $b \in [0, 2^{24} - 1]$ .

**What Diehard actually does (multiple windows)**

It repeats the same test for different **bit positions** to check bias in different bits:

Window 1: bits 1–24

Window 2: bits 2–25

...

Window 9: bits 9–32

In code form (0-indexed shift):

```
birthday = (x >> shift) & 0xFFFFF; // shift = 0..8
```

So the same file is read, and for each shift=0..8 it makes birthdays from different bits.

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**3) How many bytes your file must have (for Birthday test)**

Diehard Birthday Spacings uses:

- **$m = 512$  birthdays per trial**
- **500 trials**
- So per bit-window it consumes:

$$512 \times 500 = 256,000 \text{ uint32 numbers}$$

That is:

- **256,000 ints × 4 bytes = 1,024,000 bytes (~1.0 MB)** per window

But it runs **9 windows** (shift 0..8), so total:

- **9 × 1,024,000 = 9,216,000 bytes (~8.8 MiB)**

✓ So for the Birthday test alone, generate at least **~9.3 MB** of random bytes.

#### 4) In plain “workflow” terms

1. You generate a binary file rng.dat (raw bytes).
2. Diehard reads it as uint32 stream (4 bytes at a time).
3. For Birthday test:
  - pick 512 numbers → convert each to 24-bit birthday
  - sort → compute gaps → count repeated gaps
  - repeat 500 times
4. Repeat steps again for different bit shifts (0..8).

If you tell me **how you currently generate your byte file** (e.g., “I write bytes from C rand() / from ChaCha20 / from my own RNG”), I’ll tell you the **exact safest format** (little-endian uint32 packing) and a tiny verification method to ensure Diehard is reading it correctly.

**You said:**

now after it how i will apply chi saure

**ChatGPT said:**

Thought for 12s

In Diehard “Birthday spacings”, you first compute the **repeat count**  $K$  (how many spacing-collisions you saw). After that, you can apply **chi-square** in two easy ways:

#### Option A (easiest): Chi-square on the p-values (uniformity test)

##### Step 1: For each trial, convert $K$ to a p-value

Under a good RNG, the collision count per trial is approximately

$$K \sim \text{Poisson}(\lambda), \lambda \approx \frac{m^3}{4N}.$$

In Diehard’s standard birthday test:  $m = 512$ ,  $N = 2^{24} \Rightarrow \lambda \approx 2$ .

For each trial  $i$ , you get  $K_i$ . Turn it into a p-value using the Poisson CDF:

- upper-tail p-value:

$$p_i = \Pr(\text{Poisson}(\lambda) \geq K_i) = 1 - F_{\text{Pois}}(K_i - 1; \lambda).$$

Now if the RNG is good, these  $p_i$ 's should look **Uniform(0,1)**.

### Step 2: Bin the p-values and do chi-square

Take  $T$  trials (e.g.,  $T = 500$ ). Choose  $B$  bins (common:  $B = 10$ ):

- Bin 1: [0,0.1)
- Bin 2: [0.1,0.2)
- ...
- Bin 10: [0.9, 1.0]

Count how many p-values fall in each bin  $\rightarrow$  observed counts  $O_j$ .

Expected count in each bin:

$$E_j = \frac{T}{B}.$$

Chi-square statistic:

$$\chi^2 = \sum_{j=1}^B \frac{(O_j - E_j)^2}{E_j}.$$

Degrees of freedom:

$$\text{df} = B - 1.$$

Then compute the chi-square p-value:

$$p_{\chi^2} = \Pr(\chi_{\text{df}}^2 \geq \chi_{\text{obs}}^2).$$

✓ If  $p_{\chi^2}$  is extremely small (like  $< 10^{-6}$ ) or extremely close to 1 repeatedly, that's suspicious.

**Why this is easiest:** you don't worry about Poisson bins; you just check whether p-values are uniform.

### Option B: Chi-square directly on the Poisson counts $K$

Instead of converting to p-values, you can test whether the histogram of  $K_i$  matches a  $\text{Poisson}(\lambda)$ .

#### Step 1: Choose bins for $K$

Example bins:

- 0, 1, 2, 3, 4, and  $\geq 5$

(You want expected counts  $\geq \sim 5$  in each bin; merge the tail if needed.)

### Step 2: Compute expected counts from Poisson probabilities

For bin  $k$ :

$$E_k = T \cdot \Pr(K = k) = T \cdot e^{-\lambda} \frac{\lambda^k}{k!}.$$

Tail bin:

$$E_{\geq 5} = T \cdot \Pr(K \geq 5) = T \cdot \left(1 - \sum_{k=0}^4 e^{-\lambda} \frac{\lambda^k}{k!}\right).$$

### Step 3: Chi-square statistic

$$\chi^2 = \sum_{\text{bins}} \frac{(O-E)^2}{E}.$$

Degrees of freedom:

- If  $\lambda$  is **fixed** (like  $\lambda = 2$  from theory):  
 $\text{df} = \text{\#bins} - 1$
- If you **estimate**  $\lambda$  from data (using  $\hat{\lambda} = \bar{K}$ ):  
 $\text{df} = \text{\#bins} - 1 - 1 = \text{\#bins} - 2$

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### Which should you use?

- If you're implementing it yourself: **Option A (p-values  $\rightarrow$  chi-square uniformity)** is simplest and robust.
- If you want to stay closest to the "Poisson collision" story: **Option B** is fine.

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If you tell me what you are doing exactly:

- are you getting **one  $K$  per trial** (500 values), or
- one  $K$  per **bit-window** (9 sets of 500),  
I'll write the exact loop (and the exact  $\lambda$ ) you should use for your setup.

**You said:**

give code in c

**ChatGPT said:**

Thought for 29s

```

/* Diehard-style Birthday Spacings test (core) + Chi-square on p-values Input: raw binary byte file
(rng.dat) read as little-endian uint32 words. For each shift in [0..8], we: - run TRIALS trials - each trial
uses M uint32 words -> birthdays are 24-bit: (x >> shift) & 0xFFFFF - compute circular spacings (M
spacings), count repeated spacings => K - convert K to p-value using Poisson(lambda) - apply chi-
square uniformity test on p-values (10 bins) Build: gcc -O2 -std=c11 birthday_chi.c -lm Run: ./a.out
rng.dat */ #include <stdio.h> #include <stdlib.h> #include <stdint.h> #include <math.h> #include
<string.h> #ifndef M_PI #define M_PI 3.14159265358979323846 #endif /* -----
PARAMETERS (edit if you want) ----- */ #define M 512 /* birthdays per trial */ #define
TRIALS 500 /* number of trials */ #define NBINS 10 /* bins for chi-square on p-values */ #define N24
(1u << 24) /* range size N = 2^24 */ #define MASK24 0xFFFFFu /* ----- little-endian
uint32 read ----- */ static int read_u32_le(FILE *fp, uint32_t *out) { unsigned char b[4]; if
(fread(b, 1, 4, fp) != 4) return 0; *out = (uint32_t)b[0] | ((uint32_t)b[1] << 8) | ((uint32_t)b[2] << 16)
| ((uint32_t)b[3] << 24); return 1; } /* ----- qsort comparators ----- */ static int
cmp_u32(const void *a, const void *b) { uint32_t x = *(const uint32_t*)a, y = *(const uint32_t*)b;
return (x > y) - (x < y); } /* ----- Poisson survival: P(X >= k) ----- We compute CDF
up to k-1 by recurrence: p0 = exp(-lambda) p_{i+1} = p_i * lambda/(i+1) */ static double
poisson_sf(int k, double lambda) { if (k <= 0) return 1.0; if (lambda <= 0.0) return 0.0; double p =
exp(-lambda); /* P(X=0) */ double cdf = p; /* sum_{i=0}^0 p_i */ for (int i = 0; i < k-1; i++) { /* move
from p_i to p_{i+1} */ p *= lambda / (double)(i + 1 + 0); /* careful with indexing below */ /* The line
above is wrong indexing if left like this; do it cleanly: */ /* We'll redo properly below (keep function
correct) */ return 0.0; } /* Correct Poisson SF implementation */ static double poisson_sf_correct(int
k, double lambda) { if (k <= 0) return 1.0; double p = exp(-lambda); /* p0 */ double cdf = p; for (int i =
0; i < k-1; i++) { /* p_{i+1} from p_i */ p *= lambda / (double)(i + 1); cdf += p; /* tiny guard: if cdf is
essentially 1, sf ~ 0 */ if (cdf >= 1.0 - 1e-15) return 0.0; } double sf = 1.0 - cdf; if (sf < 0.0) sf = 0.0;
return sf; } /* ----- Chi-square p-value via regularized upper gamma Q(a,x) -----
We need p = P(ChiSq_df >= chi2) = Q(df/2, chi2/2). Implementation adapted from classic Numerical
Recipes style (series + continued fraction). */ static double gammln(double xx) { static const double
cof[6] = { 76.18009172947146, -86.50532032941677, 24.01409824083091, -1.231739572450155,
0.1208650973866179e-2, -0.5395239384953e-5 }; double x = xx, y = xx; double tmp = x + 5.5; tmp -=
(x + 0.5) * log(tmp); double ser = 1.000000000190015; for (int j = 0; j < 6; j++) ser += cof[j] / ++y;
return -tmp + log(2.5066282746310005 * ser / x); } static void gser(double *gamser, double a, double
x, double *gln) { const int ITMAX = 200; const double EPS = 3.0e-14; *gln = gammln(a); if (x <= 0.0) {
*gamser = 0.0; return; } double sum = 1.0 / a; double del = sum; double ap = a; for (int n = 1; n <=
ITMAX; n++) { ap += 1.0; del *= x / ap; sum += del; if (fabs(del) < fabs(sum) * EPS) break; } *gamser =
sum * exp(-x + a * log(x) - (*gln)); } static void gcf(double *gammcf, double a, double x, double *gln) {
const int ITMAX = 200; const double EPS = 3.0e-14; const double FPMIN = 1.0e-300; *gln =
gammln(a); double b = x + 1.0 - a; double c = 1.0 / FPMIN; double d = 1.0 / b; double h = d; for (int i =
1; i <= ITMAX; i++) { double an = -i * (i - a); b += 2.0; d = an * d + b; if (fabs(d) < FPMIN) d = FPMIN; c =
b + an / c; if (fabs(c) < FPMIN) c = FPMIN; d = 1.0 / d; double del = d * c; h *= del; if (fabs(del - 1.0) <
EPS) break; } *gammcf = exp(-x + a * log(x) - (*gln)) * h; } /* Regularized upper incomplete gamma
Q(a,x) */ static double gammq(double a, double x) { if (x < 0.0 || a <= 0.0) return 0.0; double gamser,
gammcf, gln; if (x < a + 1.0) { gser(&gamser, a, x, &gln); return 1.0 - gamser; /* Q = 1 - P */ } else {
gcf(&gammcf, a, x, &gln); return gammcf; /* Q directly */ } } /* ----- Birthday spacings: one
shift ----- */ static int birthday_one_shift(FILE *fp, int shift, double lambda, int
bins[NBINS], double *meanK_out) { uint32_t birthdays[M]; uint32_t spacings[M]; memset(bins, 0,
NBINS * sizeof(int)); long long sumK = 0; for (int t = 0; t < TRIALS; t++) { /* read M uint32 -> birthdays
*/ for (int i = 0; i < M; i++) { uint32_t x; if (!read_u32_le(fp, &x)) { fprintf(stderr, "EOF: need more data
(shift=%d, trial=%d, i=%d)\n", shift, t, i); return 0; } birthdays[i] = (x >> shift) & MASK24; } /* sort

```

```

birthdays */ qsort(birthdays, M, sizeof(uint32_t), cmp_u32); /* circular spacings (M of them) */ for
(int i = 0; i < M - 1; i++) spacings[i] = birthdays[i + 1] - birthdays[i]; spacings[M - 1] = (birthdays[0] +
N24) - birthdays[M - 1]; /* sort spacings */ qsort(spacings, M, sizeof(uint32_t), cmp_u32); /* count
repeated spacings */ int K = 0; for (int i = 1; i < M; i++) { if (spacings[i] == spacings[i - 1]) K++; } sumK
+= K; /* Poisson p-value: upper tail  $P(X \geq K)$  */ double p = poisson_sf_correct(K, lambda); /* bin p
into 10 equal bins */ int b = (int)floor(p * NBINS); if (b < 0) b = 0; if (b >= NBINS) b = NBINS - 1;
bins[b]++; } *meanK_out = (double)sumK / (double)TRIALS; return 1; } static void
chi_square_uniform(const int bins[NBINS], int n, double *chi2_out, double *p_out) { double E =
(double)n / (double)NBINS; double chi2 = 0.0; for (int i = 0; i < NBINS; i++) { double O =
(double)bins[i]; double d = O - E; chi2 += (d * d) / E; } int df = NBINS - 1; double p = gammq(0.5 * df,
0.5 * chi2); /* upper-tail */ *chi2_out = chi2; *p_out = p; } int main(int argc, char **argv) { if (argc <
2) { fprintf(stderr, "Usage: %s rng.dat\n", argv[0]); return 1; } FILE *fp = fopen(argv[1], "rb"); if (!fp) {
perror("fopen"); return 1; } /* Diehard lambda approximation:  $\lambda \approx m^3/(4N)$ , with  $N=2^{24}$  */
double lambda = ((double)M * (double)M * (double)M) / (4.0 * (double)N24); printf("Birthday
Spacings test (M=%d, TRIALS=%d, N=2^24), lambda≈%.6f\n", M, TRIALS, lambda); printf("Input is
read as little-endian uint32 from the byte file.\n\n"); for (int shift = 0; shift <= 8; shift++) { rewind(fp);
int bins[NBINS]; double meanK = 0.0; if (!birthday_one_shift(fp, shift, lambda, bins, &meanK)) {
fclose(fp); return 2; } double chi2, pchi; chi_square_uniform(bins, TRIALS, &chi2, &pchi);
printf("Shift=%d (using bits %d..%d for 24-bit birthdays):\n", shift, shift+1, shift+24); printf("
mean(K)=%.4f\n", meanK); printf(" p-value bins (10 equal bins): "); for (int i = 0; i < NBINS; i++)
printf("%d%s", bins[i], (i==NBINS-1) ? "" : ","); printf("\n"); printf(" ChiSq=%.4f, df=%d, ChiSq-
p=%.6g\n\n", chi2, NBINS-1, pchi); } fclose(fp); return 0; }

```

### What you do with it

1. Generate a **raw binary** file rng.dat (just bytes).
2. Compile & run:

```
gcc -O2 -std=c11 birthday_chi.c -lm ./a.out rng.dat
```

### How big your file should be (minimum)

For **one shift** it needs  $M \cdot \text{TRIALS} = 512 \cdot 500 = 256000$  uint32 numbers  $\Rightarrow 256000 \cdot 4 = 1,024,000$  bytes (~1 MB).

Since the code **rewinds** and reuses the same data for shifts 0..8, **~1 MB is enough** to run all shifts here.

If you want the “more strict” approach (different fresh data for each shift), tell me — I’ll modify the code to *not rewind* and then you’ll need ~9 MB.

### You said:

```

#include "header.h" static real chi_fit; static int dgf; int ucmpr(const void *u1, const void *u2);
/*compute the statistic of goodness of fit to a Poisson distribution*/ real P_fit(real lambda, counter
*obs, int no_obs) { counter *f, dim=no_obs/5; register int i=-1; register counter j, k=0; real
rest=no_obs, *Ef; f=(counter*)calloc(dim, sizeof(counter)); Ef=(real*)calloc(dim, sizeof(real));
qsort(obs, no_obs, sizeof(counter), ucmpr); for(j=0; j<dim; ++j){ while( Ef[j]<5 ){ ++i;
Ef[j]+=no_obs*Poisson(lambda, i); } while( obs[k]<=i ){ ++f[j]; ++k; } rest-=Ef[j]; if( rest<5 ){ Ef[j]+=rest;
f[j]+=no_obs-k; break; } } chi_fit=0; for(i=0; i<=j; ++i){ chi_fit+=(f[i]-Ef[i])*(f[i]-Ef[i])/Ef[i]; } dgf=j; /*
cfree(f, dim, sizeof(counter)); cfree(Ef, dim, sizeof(real)); */ return 1-Chisq(dgf,chi_fit); } void
bday(char *filename) { const counter no_obs=500, no_bday=1024, no_bits=24; register const

```

```

counter mask=pow(2,no_bits)-1; const real lambda=pow(no_bday,3)/(4*pow(2,no_bits)); register int
rt; register int i, k, sum; uniform *bdspace; counter no_dup, *obs; real pvalue, *p; puts("\n\t|-----
-----|"); puts("\t| This is the BIRTHDAY SPACINGS TEST |");
puts("\t| Choose m birthdays in a \"year\" of n days. List the spacings |"); puts("\t| between the
birthdays. Let j be the number of values that |"); puts("\t| occur more than once in that list, then j is
asymptotically |"); puts("\t| Poisson distributed with mean m^3/(4n). Experience shows n |");
puts("\t| must be quite large, say n>=2^18, for comparing the results |"); puts("\t| to the Poisson
distribution with that mean. This test uses |"); puts("\t| n=2^24 and m=2^10, so that the underlying
distribution for j |"); puts("\t| is taken to be Poisson with lambda=2^30/(2^26)=16. A sample |");
puts("\t| of 200 j's is taken, and a chi-square goodness of fit test |"); puts("\t| provides a p value. The
first test uses bits 1-24 (counting |"); puts("\t| from the left) from integers in the specified file. Then
the |"); puts("\t| file is closed and reopened, then bits 2-25 of the same inte- |"); puts("\t| gers are
used to provide birthdays, and so on to bits 9-32. |"); puts("\t| Each set of bits provides a p-value,
and the nine p-values |"); puts("\t| provide a sample for a KSTEST. |"); puts("\t|-----
-----|"); printf("\t\tRESULTS OF BIRTHDAY SPACINGS TEST FOR %s\n",
filename); printf("\t(no_bdays=%lu, no_days/yr=2^%lu,\"no_bday, no_bits); printf(" lambda=%f,
sample size=%lu)\n\n", lambda, no_obs); printf("\tBits used\tmean\t\tchisqr\t\t p-value\n");
obs=(counter*)malloc(no_obs*sizeof(counter)); p=(real*)malloc((32-no_bits+1)*sizeof(real));
bdspace=(uniform*)malloc(no_bday*sizeof(uniform)); #define GETDAY ( (uni(filename) >> rt) & mask
) for(rt=32-no_bits; rt>=0; --rt){ sum=0; for(k=0; k<no_obs; ++k){ for(i=0; i<no_bday; ++i){
bdspace[i]=GETDAY; } qsort(bdspace, no_bday, sizeof(uniform), ucmp); for(i=no_bday-1; i>=1; --i){
bdspace[i]-=bdspace[i-1]; } qsort(bdspace, no_bday, sizeof(uniform), ucmp); no_dup=0; for(i=1;
i<no_bday; ++i){ if(bdspace[i]!=bdspace[i-1]) continue; else ++no_dup; } sum+=no_dup;
obs[k]=no_dup; } uni("close"); p[rt]=P_fit(lambda, obs, no_obs); printf("\t %lu to %d", 32-no_bits-
rt,32-rt); printf("\t%.2f\t%.4f\t\t%f\n", (real)sum/no_obs, chi_fit, p[rt]); } /*clean up*/
free(bdspace); free(obs); pvalue=KStest(p,32-no_bits+1); printf("\n\t\t\tdegree of freedoms is:
%d\n", dgf); puts("\t|-----"); printf("\t\t p-value for KStest
on those %lu p-values: %f",32-no_bits+1,pvalue); puts("\n"); free(p); return; } /*main() { char ch;
bday("binc"); return; }*/ is the code is same u give

```

**ChatGPT said:**

Thought for 17s

It's the **same Diehard Birthday Spacings** idea, but it is **not the same as the C code I gave you**. Your code is basically the **original Diehard-style implementation** (Poisson GOF + KS over 9 bit-windows). Mine was a **simplified core test + chi-square on p-values**.

**What is the same (conceptually)**

- Read random data as **32-bit integers** (uni(filename) in your code; my code reads 4 bytes → uint32\_t).
- Form 24-bit “birthdays” using **bit windows**:

$$\text{day} = (x \gg rt) \& 0\text{FFFFFF}$$

- For each window, repeat many trials, compute “duplicate spacings”, turn it into a p-value.



## Key differences

### 1) m (number of birthdays per trial)

- **Your code:** no\_bday = 1024 (=  $2^{10}$ )
- **My code:** M = 512

So the test strength and lambda change.

### 2) $\lambda$ (Poisson mean)

- **Your code:**

$$\lambda = \frac{1024^3}{4 \cdot 2^{24}} = 16$$

- **My code:**

$$\lambda = \frac{512^3}{4 \cdot 2^{24}} \approx 2$$

So even if everything else matched, the expected collision counts differ a lot.

### 3) How chi-square is applied

- **Your code:** does a **chi-square goodness-of-fit** test of the *distribution* of  $\{K_i\}$  to **Poisson( $\lambda$ )** via P\_fit().
  - It bins Poisson probabilities until expected count  $\geq 5$ , then computes:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- **My code:** I did **chi-square uniformity** on the **p-values** (10 equal bins on [0,1]) — simpler, not the original Diehard method.

### 4) KS test stage

- **Your code:** After getting **9 p-values** (for bit windows), it runs **KStest(p, 9)**.
- **My code:** I did **no KS across shifts** (I only printed chi-square p per shift).

### 5) Data reuse across windows

- **Your code:** closes/reopens the RNG (uni("close")) each window (i.e., re-reads the same file again per window).
- **My code:** used rewind(fp) (same effect).

---

## Important possible issue in your posted code (check this!)

In birthday spacings, you normally need the **wrap-around spacing** too:

$$s_{m-1} = (x_0 + N) - x_{m-1}$$

In your loop:

```
for(i=no_bday-1; i>=1; --i){ bdspace[i]=bdspace[i-1]; }
```

bdspace[0] is left as the **smallest birthday**, not the wrap-around spacing.

Unless uniform/uni()/your original library does something special elsewhere, this looks like a **missing circular spacing** and can change the statistics.

✓ In my code I explicitly did:

```
spacings[M-1] = (birthdays[0] + N24) - birthdays[M-1];
```

### Bottom line

- **Same test idea:** yes.
- **Same code as I gave:** no (different m, different lambda, different chi-square method, and your code includes KS step).
- Also **verify wrap-around spacing**, because that's a common (and serious) bug.

If you want, paste your uni() and uniform/counter typedefs (or header.h parts), and I'll tell you **exactly** whether your spacing computation is correct for your implementation.

**You said:**

so u give coorect code in c everything is clear and i have file outputdat

**ChatGPT said:**

Thought for 18s

Below is a **correct, self-contained C program** that implements the **full Diehard "Birthday Spacings" test** the standard way:

- Reads your **raw byte file** (output.dat) as **little-endian uint32**.
- Uses **n = 2^24, m = 2^10 = 1024 birthdays per sample, 500 samples**.
- For each **24-bit window** (bits 1–24, 2–25, ..., 9–32 ⇒ 9 windows):
  - generates 500 values of **K = (# duplicated spacings)** (with **wrap-around spacing included** ✓)
  - computes a **chi-square goodness-of-fit** to **Poisson(λ=16)**
  - prints **mean K, chi-square, p-value**
- Then performs a **KS test** on the 9 p-values and prints the KS p-value.

**Minimum file size needed** (because we rewind and reuse the same data for each window, like Diehard):

1024 × 500 uint32 = 512,000 uint32 = **2,048,000 bytes (~2 MB)**.

If your file is bigger, that's fine.

---

```
/* Diehard Birthday Spacings Test (full style): - n = 2^24 days - m = 1024 birthdays per sample - 500
samples per bit-window - 9 windows: bits 1-24 ... 9-32 (i.e., shifts 8..0 from MSB view) - For each
window: chi-square GOF of K to Poisson(lambda=16) - KS test on the 9 resulting p-values Build: gcc -
O2 -std=c11 diehard_bday.c -lm Run: ./a.out output.dat */ #include <stdio.h> #include <stdlib.h>
#include <stdint.h> #include <math.h> #include <string.h> /* ----- Parameters (Diehard
standard for this test) ----- */ #define NO_OBS 500 #define NO_BDAY 1024 #define NO_BITS
24 #define N_DAYS (1u << 24) #define MASK24 0xFFFFFu /* ----- Little-endian uint32 reader
from raw byte file ----- */ static int read_u32_le(FILE *fp, uint32_t *out) { unsigned char b[4];
if (fread(b, 1, 4, fp) != 4) return 0; *out = (uint32_t)b[0] | ((uint32_t)b[1] << 8) | ((uint32_t)b[2] <<
16) | ((uint32_t)b[3] << 24); return 1; } /* ----- qsort comparator ----- */ static int
cmp_u32(const void *a, const void *b) { uint32_t x = *(const uint32_t*)a; uint32_t y = *(const
uint32_t*)b; return (x > y) - (x < y); } /* ----- log-gamma helpers for chi-square p-value -----
----- We need upper-tail p = P(ChiSq_df >= x) = Q(df/2, x/2). Implemented via regularized upper
incomplete gamma Q(a,x). */ static double gammaln(double xx) { static const double cof[6] = {
76.18009172947146, -86.50532032941677, 24.01409824083091, -1.231739572450155,
0.1208650973866179e-2, -0.5395239384953e-5 }; double x = xx, y = xx; double tmp = x + 5.5; tmp -=
(x + 0.5) * log(tmp); double ser = 1.000000000190015; for (int j = 0; j < 6; j++) ser += cof[j] / ++y;
return -tmp + log(2.5066282746310005 * ser / x); } static void gser(double *gamser, double a, double
x, double *gln) { const int ITMAX = 200; const double EPS = 3.0e-14; *gln = gammaln(a); if (x <= 0.0) {
*gamser = 0.0; return; } double sum = 1.0 / a; double del = sum; double ap = a; for (int n = 1; n <=
ITMAX; n++) { ap += 1.0; del *= x / ap; sum += del; if (fabs(del) < fabs(sum) * EPS) break; } *gamser =
sum * exp(-x + a * log(x) - (*gln)); } static void gcf(double *gammcf, double a, double x, double *gln) {
const int ITMAX = 200; const double EPS = 3.0e-14; const double FPMIN = 1.0e-300; *gln =
gammaln(a); double b = x + 1.0 - a; double c = 1.0 / FPMIN; double d = 1.0 / b; double h = d; for (int i =
1; i <= ITMAX; i++) { double an = -i * (i - a); b += 2.0; d = an * d + b; if (fabs(d) < FPMIN) d = FPMIN; c =
b + an / c; if (fabs(c) < FPMIN) c = FPMIN; d = 1.0 / d; double del = d * c; h *= del; if (fabs(del - 1.0) <
EPS) break; } *gammcf = exp(-x + a * log(x) - (*gln)) * h; } static double gammq(double a, double x) { if
(x < 0.0 || a <= 0.0) return 0.0; double gamser, gammcf, gln; if (x < a + 1.0) { gser(&gamser, a, x, &gln);
return 1.0 - gamser; } else { gcf(&gammcf, a, x, &gln); return gammcf; } } /* Upper-tail chi-square p-
value */ static double chisq_sf(int df, double chi2) { return gammq(0.5 * df, 0.5 * chi2); } /* -----
--- Poisson pmf via recurrence (stable for small/medium k) ----- */ static double
poisson_pmf(int k, double lambda) { /* compute e^-lambda * lambda^k / k! using recurrence from 0
*/ double p0 = exp(-lambda); if (k == 0) return p0; double p = p0; for (int i = 1; i <= k; i++) { p *=
lambda / (double)i; } return p; } /* ----- Chi-square GOF of observed K's to Poisson(lambda) --
----- We follow the Diehard-style rule: combine categories so each expected >= 5. Inputs:
obs[NO_OBS] = K values (duplicates per sample) Outputs: chi2, df, p-value. */ static void
poisson_chisq_gof(const int *obs, int n, double lambda, double *chi2_out, int *df_out, double
*p_out) { /* Make a sorted copy of obs */ int *x = (int*)malloc((size_t)n * sizeof(int)); if (!x) {
fprintf(stderr, "malloc failed\n"); exit(1); } memcpy(x, obs, (size_t)n * sizeof(int)); /* sort */ /* simple
qsort comparator for int */ int cmp_int(const void *a, const void *b) { int A = *(const int*)a, B =
*(const int*)b; return (A > B) - (A < B); } qsort(x, (size_t)n, sizeof(int), cmp_int); /* Build bins by
increasing k, accumulating expected until >=5 */ double chi2 = 0.0; int bins = 0; int idx = 0; /* how
many obs consumed */ int k = -1; /* current Poisson k */ double exp_in_bin = 0.0; int obs_in_bin = 0;
while (idx < n) { /* extend bin until expected >= 5 */ while (exp_in_bin < 5.0) { k++; exp_in_bin +=
```

```
(double)n * poisson_pmf(k, lambda); /* add all observations equal to k */ while (idx < n && x[idx] <=
k) { obs_in_bin++; idx++; } /* if we are near the end, dump the tail into this bin */ double
remaining_expected = (double)n - (double)(bins == 0 ? 0 : 0); /* not used */
(void)remaining_expected; if (idx >= n) break; } /* If remaining expected mass is < 5, merge rest into
this bin: easiest way: if remaining observations are few, keep merging. */ /* We handle this by
checking if remaining expected (tail) would be <5: approximate tail expected = n*(1 - sum_{i=0..k}
pmf(i)). We'll compute tail cdf roughly by summing pmf up to k. */ /* For simplicity & correctness: if
observations remain but we are close, we'll continue making bins normally; expected will not stay
tiny with lambda=16, n=500. */ /* finalize this bin */ if (exp_in_bin > 0.0) { double diff =
(double)obs_in_bin - exp_in_bin; chi2 += (diff * diff) / exp_in_bin; bins++; } /* reset bin counters */
exp_in_bin = 0.0; obs_in_bin = 0; } /* degrees of freedom = bins - 1 (lambda fixed) */ int df = bins - 1;
if (df < 1) df = 1; double p = chisq_sf(df, chi2); free(x); *chi2_out = chi2; *df_out = df; *p_out = p; } /*
----- One-sample KS test for Uniform(0,1) ----- */ static double
ks_pvalue_uniform(const double *pvals, int n) { double *x = (double*)malloc((size_t)n *
sizeof(double)); if (!x) { fprintf(stderr, "malloc failed\n"); exit(1); } memcpy(x, pvals, (size_t)n *
sizeof(double)); int cmp_double(const void *a, const void *b) { double A = *(const double*)a, B =
*(const double*)b; return (A > B) - (A < B); } qsort(x, (size_t)n, sizeof(double), cmp_double); double D
= 0.0; for (int i = 0; i < n; i++) { double xi = x[i]; double cdf_emp_hi = (double)(i + 1) / (double)n;
double cdf_emp_lo = (double)i / (double)n; double d1 = fabs(cdf_emp_hi - xi); double d2 = fabs(xi -
cdf_emp_lo); if (d1 > D) D = d1; if (d2 > D) D = d2; } free(x); /* Kolmogorov asymptotic p-value:  $p \approx 2 \sum (-1)^{k-1} \exp(-2 k^2 D^2 n)$  */ double nn = (double)n; double z = D * sqrt(nn); /* Use the exact series
in terms of D and n */ double p = 0.0; for (int k = 1; k <= 200; k++) { double term = exp(-2.0 *
(double)k * (double)k * D * D * nn); if (k % 2 == 1) p += 2.0 * term; else p -= 2.0 * term; if (term < 1e-
12) break; } if (p < 0.0) p = 0.0; if (p > 1.0) p = 1.0; (void)z; return p; } /* ----- Main Birthday
Spacings per bit-window ----- */ static void run_bday_window(FILE *fp, int shift, double
lambda, double *meanK_out, double *chi2_out, int *df_out, double *p_out) { uint32_t
bday[NO_BDAY]; uint32_t space[NO_BDAY]; int obs[NO_OBS]; long long sumK = 0; for (int t = 0; t <
NO_OBS; t++) { for (int i = 0; i < NO_BDAY; i++) { uint32_t x; if (!read_u32_le(fp, &x)) { fprintf(stderr,
"EOF: not enough data for shift=%d (trial=%d, i=%d)\n", shift, t, i); exit(2); } bday[i] = (x >> shift) &
MASK24; } qsort(bday, NO_BDAY, sizeof(uint32_t), cmp_u32); /* Compute spacings INCLUDING wrap-around
spacing (critical) */ for (int i = 0; i < NO_BDAY - 1; i++) { space[i] = bday[i + 1] - bday[i]; }
space[NO_BDAY - 1] = (bday[0] + N_DAYS) - bday[NO_BDAY - 1]; qsort(space, NO_BDAY,
sizeof(uint32_t), cmp_u32); /* Count duplicates (total repeats) */ int K = 0; for (int i = 1; i < NO_BDAY;
i++) { if (space[i] == space[i - 1]) K++; } obs[t] = K; sumK += K; } *meanK_out = (double)sumK /
(double)NO_OBS; /* Chi-square goodness-of-fit of obs[] to Poisson(lambda) */
poisson_chisq_gof(obs, NO_OBS, lambda, chi2_out, df_out, p_out); } int main(int argc, char **argv) {
if (argc < 2) { fprintf(stderr, "Usage: %s output.dat\n", argv[0]); return 1; } const char *fname =
argv[1]; FILE *fp = fopen(fname, "rb"); if (!fp) { perror("fopen"); return 1; } /* lambda = m^3/(4n)
with m=1024, n=2^24 => 16 */ double lambda = (double)NO_BDAY * (double)NO_BDAY *
(double)NO_BDAY / (4.0 * (double)N_DAYS); printf("\nDiehard Birthday Spacings Test on file: %s\n",
fname); printf("n = 2^%d = %u, m = %d, samples = %d, lambda = %.6f\n", NO_BITS, N_DAYS,
NO_BDAY, NO_OBS, lambda); printf("Bits used (from MSB view)\t\tmean(K)\t\t\tchisq\t\t\t\t\t\t\t\t\t\t\t\t\tp-
value\n"); double pvals[9]; int idx = 0; /* Diehard uses bits 1-24, 2-25, ... 9-32. With uint32 x, that
corresponds to shifts 8 down to 0 if you imagine MSB-indexing. But mathematically it is just 9
consecutive 24-bit windows. We implement shift = 0.8 (LSB upward). This is standard in many
implementations. If you want "from the left" exactly, reverse the shift order—KS result is the same
set. */ for (int shift = 0; shift <= 8; shift++) { rewind(fp); double meanK, chi2, p; int df;
run_bday_window(fp, shift, lambda, &meanK, &chi2, &df, &p); /* report bits (1-based human
```

```
display) */ int bit_lo = shift + 1; int bit_hi = shift + NO_BITS; printf("%2d to
%2d\t\t\t%.3f\t\t%.4f\t\t%.5d\t\t%.6g\n", bit_lo, bit_hi, meanK, chi2, df, p); pvals[idx++] = p; }
fclose(fp); /* KS test on the 9 p-values */ double ks_p = ks_pvalue_uniform(pvals, 9); printf("\nKS test
on the 9 p-values: KS p-value = %.6g\n\n", ks_p); return 0; }
```

## How to run

```
gcc -O2 -std=c11 diehard_bday.c -lm ./a.out outputdat
```

## Notes (important)

- Your file **must be raw bytes** (not ASCII numbers).
- This code assumes **little-endian** packing for uint32 (most common).
- The **wrap-around spacing** is included (this is where many “almost-diehard” codes go wrong).

If you want, paste **how you generated** outputdat (did you write bytes directly, or printed integers as text?), and I’ll tell you immediately whether you should read as **binary** or **ASCII** and adjust the reader accordingly.

## You said:

```
#include "header.h" static real chi_fit; static int dgf; int ucmpr(const void *u1, const void *u2);
/*compute the statistic of goodness of fit to a Poisson distribution*/ real P_fit(real lambda, counter
*obs, int no_obs) { counter *f, dim=no_obs/5; register int i=-1; register counter j, k=0; real
rest=no_obs, *Ef; f=(counter*)calloc(dim, sizeof(counter)); Ef=(real*)calloc(dim, sizeof(real));
qsort(obs, no_obs, sizeof(counter), ucmpr); for(j=0; j<dim; ++j){ while( Ef[j]<5 ){ ++i;
Ef[j]+=no_obs*Poisson(lambda, i); } while( obs[k]<=i ){ ++f[j]; ++k; } rest-=Ef[j]; if( rest<5 ){ Ef[j]+=rest;
f[j]+=no_obs-k; break; } } chi_fit=0; for(i=0; i<=j; ++i){ chi_fit+=(f[i]-Ef[i])*(f[i]-Ef[i])/Ef[i]; } dgf=j; /*
cfree(f, dim, sizeof(counter)); cfree(Ef, dim, sizeof(real)); */ return 1-Chisq(dgf,chi_fit); } void
bday(char *filename) { const counter no_obs=500, no_bday=1024, no_bits=24; register const
counter mask=pow(2,no_bits)-1; const real lambda=pow(no_bday,3)/(4*pow(2,no_bits)); register int
rt; register int i, k, sum; uniform *bdspace; counter no_dup, *obs; real pvalue, *p; puts("\n\t|-----
-----|"); puts("\t| This is the BIRTHDAY SPACINGS TEST |");
puts("\t|Choose m birthdays in a \"year\" of n days. List the spacings |"); puts("\t|between the
birthdays. Let j be the number of values that |"); puts("\t|occur more than once in that list, then j is
asymptotically |"); puts("\t|Poisson distributed with mean m^3/(4n). Experience shows n |");
puts("\t|must be quite large, say n>=2^18, for comparing the results |"); puts("\t|to the Poisson
distribution with that mean. This test uses |"); puts("\t|n=2^24 and m=2^10, so that the underlying
distribution for j |"); puts("\t|is taken to be Poisson with lambda=2^30/(2^26)=16. A sample |");
puts("\t|of 200 j's is taken, and a chi-square goodness of fit test |"); puts("\t|provides a p value. The
first test uses bits 1-24 (counting |"); puts("\t|from the left) from integers in the specified file. Then
the |"); puts("\t|file is closed and reopened, then bits 2-25 of the same inte- |"); puts("\t|gers are
used to provide birthdays, and so on to bits 9-32. |"); puts("\t|Each set of bits provides a p-value,
and the nine p-values |"); puts("\t|provide a sample for a KSTEST. |"); puts("\t|-----
-----|"); printf("\t\tRESULTS OF BIRTHDAY SPACINGS TEST FOR %s\n",
filename); printf("\t(no_bdays=%lu, no_days/yr=2^%lu,\"no_bday, no_bits); printf(" lambda=%2f,
sample size=%lu)\n\n", lambda, no_obs); printf("\tBits used\ttmean\t\tchisqr\t\tp-value\n");
obs=(counter*)malloc(no_obs*sizeof(counter)); p=(real*)malloc((32-no_bits+1)*sizeof(real));
bdspace=(uniform*)malloc(no_bday*sizeof(uniform)); #define GETDAY ( (uni(filename) >> rt) & mask
) for(rt=32-no_bits; rt>=0; --rt){ sum=0; for(k=0; k<no_obs; ++k){ for(i=0; i<no_bday; ++i){
```

```

bdspace[i]=GETDAY; } qsort(bdspace, no_bday, sizeof(uniform), ucmp); for(i=no_bday-1; i>=1; --i){
bdspace[i]=bdspace[i-1]; } qsort(bdspace, no_bday, sizeof(uniform), ucmp); no_dup=0; for(i=1;
i<no_bday; ++i){ if(bdspace[i]!=bdspace[i-1]) continue; else ++no_dup; } sum+=no_dup;
obs[k]=no_dup; } uni("close"); p[rt]=P_fit(lambda, obs, no_obs); printf("\t %lu to %d", 33-no_bits-
rt,32-rt); printf("\t%.2f\t%.4f\t%.4f\n", (real)sum/no_obs, chi_fit, p[rt]); } /*clean up*/
free(bdspace); free(obs); pvalue=KStest(p,32-no_bits+1); printf("\n\t\tdegree of freedoms is:
%d\n", dgf); puts("\t-----"); printf("\t\t p-value for KStest
on those %lu p-values: %f",32-no_bits+1,pvalue); puts("\n"); free(p); return; } /*main() { char ch;
bday("binc"); return; }*/ give this code in beeter way

```

### ChatGPT said:

Here's your same code rewritten in a **clean, correct, self-contained, standard C style** (no header.h, no uni(), no custom typedefs), while keeping the **exact Diehard logic**:

#### ✓ Improvements vs your version

- Uses **normal C types** (uint32\_t, int, double)
- Reads your **binary byte file** directly (little-endian uint32\_t)
- Correctly includes the **wrap-around spacing** (your version likely misses it)
- No macros that hide I/O
- No memory leaks
- Clear separation: birthday\_test\_window(), poisson\_chisq\_gof(), ks\_uniform()
- Prints the same table + KS result

### Correct + cleaner Diehard Birthday Spacings (C)

```

/* Clean Diehard BIRTHDAY SPACINGS test Parameters (classic Diehard): n = 2^24 days m = 1024
birthdays per sample no_obs = 500 samples per bit-window windows = 9 (24-bit windows across 32-
bit words) For each window: - compute K (duplicate spacings) for each sample - do chi-square GOF to
Poisson(lambda = m^3/(4n) = 16) - output p-value Then: - KS test on the 9 p-values Build: gcc -O2 -
std=c11 bday_clean.c -lm Run: ./a.out output.dat */ #include <stdio.h> #include <stdlib.h> #include
<stdint.h> #include <string.h> #include <math.h> #define NO_OBS 500 #define NO_BDAY 1024
#define NO_BITS 24 #define N_DAYS (1u << 24) #define MASK24 0xFFFFFu #define WINDOWS (32 -
NO_BITS + 1) /* 9 */ static int cmp_u32(const void *a, const void *b) { uint32_t x = *(const
uint32_t*)a, y = *(const uint32_t*)b; return (x > y) - (x < y); } static int cmp_d(const void *a, const
void *b) { double x = *(const double*)a, y = *(const double*)b; return (x > y) - (x < y); } static int
cmp_i(const void *a, const void *b) { int x = *(const int*)a, y = *(const int*)b; return (x > y) - (x < y); }
/* ----- read one little-endian uint32 from raw byte file ----- */ static int read_u32_le(FILE *fp,
uint32_t *out) { unsigned char b[4]; if (fread(b, 1, 4, fp) != 4) return 0; *out = (uint32_t)b[0] |
((uint32_t)b[1] << 8) | ((uint32_t)b[2] << 16) | ((uint32_t)b[3] << 24); return 1; } /* ----- Regularized
upper incomplete gamma Q(a,x) for chi-square SF ----- */ /* (Numerical Recipes style
implementation; stable and standard.) */ static double gammln(double xx) { static const double
cof[6] = { 76.18009172947146, -86.50532032941677, 24.01409824083091, -1.231739572450155,
0.1208650973866179e-2, -0.5395239384953e-5 }; double x = xx, y = xx; double tmp = x + 5.5; tmp -=

```

```

(x + 0.5) * log(tmp); double ser = 1.000000000190015; for (int j = 0; j < 6; j++) ser += cof[j] / ++y;
return -tmp + log(2.5066282746310005 * ser / x); } static void gser(double *gamser, double a, double
x, double *gln) { const int ITMAX = 200; const double EPS = 3.0e-14; *gln = gammln(a); if (x <= 0.0) {
*gamser = 0.0; return; } double sum = 1.0 / a, del = sum, ap = a; for (int n = 1; n <= ITMAX; n++) { ap
+= 1.0; del *= x / ap; sum += del; if (fabs(del) < fabs(sum) * EPS) break; } *gamser = sum * exp(-x + a *
log(x) - (*gln)); } static void gcf(double *gammcf, double a, double x, double *gln) { const int ITMAX =
200; const double EPS = 3.0e-14; const double FPMIN = 1.0e-300; *gln = gammln(a); double b = x +
1.0 - a; double c = 1.0 / FPMIN; double d = 1.0 / b; double h = d; for (int i = 1; i <= ITMAX; i++) {
double an = -i * (i - a); b += 2.0; d = an * d + b; if (fabs(d) < FPMIN) d = FPMIN; c = b + an / c; if (fabs(c)
< FPMIN) c = FPMIN; d = 1.0 / d; double del = d * c; h *= del; if (fabs(del - 1.0) < EPS) break; }
*gammcf = exp(-x + a * log(x) - (*gln)) * h; } static double gammq(double a, double x) { if (x < 0.0 || a
<= 0.0) return 0.0; double gamser, gammcf, gln; if (x < a + 1.0) { gser(&gamser, a, x, &gln); return 1.0 -
gamser; } else { gcf(&gammcf, a, x, &gln); return gammcf; } } static double chisq_sf(int df, double
chi2) { return gammq(0.5 * df, 0.5 * chi2); } /* ----- Poisson pmf by recurrence: p(k) = e^-λ λ^k / k! -
----- */ static double poisson_pmf(int k, double lambda) { double p = exp(-lambda); for (int i = 1; i <=
k; i++) p *= lambda / (double)i; return p; } /* ----- Chi-square GOF to Poisson(lambda), bins
combined so expected >= 5 ----- Returns p-value, and outputs chi2 and df. */ static double
poisson_chisq_gof(const int *obs, int n, double lambda, double *chi2_out, int *df_out) { int *x =
(int*)malloc((size_t)n * sizeof(int)); if (!x) { fprintf(stderr, "malloc failed\n"); exit(1); } memcpy(x, obs,
(size_t)n * sizeof(int)); qsort(x, (size_t)n, sizeof(int), cmp_i); double chi2 = 0.0; int bins = 0; int idx = 0;
int k = -1; while (idx < n) { double E = 0.0; int O = 0; /* grow a bin until E >= 5 */ while (E < 5.0) { k++; E
+= (double)n * poisson_pmf(k, lambda); while (idx < n && x[idx] <= k) { O++; idx++; } /* if we already
consumed all obs, stop */ if (idx >= n) break; } if (E > 0.0) { double d = (double)O - E; chi2 += (d * d) /
E; bins++; } } free(x); int df = bins - 1; /* lambda fixed (not estimated) */ if (df < 1) df = 1; *chi2_out =
chi2; *df_out = df; return chisq_sf(df, chi2); } /* ----- 1-sample KS test against Uniform(0,1),
asymptotic p-value ----- */ static double ks_uniform_pvalue(const double *pvals, int n) { double *x
= (double*)malloc((size_t)n * sizeof(double)); if (!x) { fprintf(stderr, "malloc failed\n"); exit(1); }
memcpy(x, pvals, (size_t)n * sizeof(double)); qsort(x, (size_t)n, sizeof(double), cmp_d); double D =
0.0; for (int i = 0; i < n; i++) { double xi = x[i]; double F_hi = (double)(i + 1) / (double)n; double F_lo =
(double)i / (double)n; double d1 = fabs(F_hi - xi); double d2 = fabs(xi - F_lo); if (d1 > D) D = d1; if (d2
> D) D = d2; } free(x); /* Kolmogorov asymptotic series */ double nn = (double)n; double p = 0.0; for
(int k = 1; k <= 200; k++) { double term = exp(-2.0 * (double)k * (double)k * D * D * nn); p += (k % 2 ?
2.0 : -2.0) * term; if (term < 1e-12) break; } if (p < 0.0) p = 0.0; if (p > 1.0) p = 1.0; return p; } /* -----
compute one window: returns p-value for GOF, plus meanK, chi2, df ----- */ static void
birthday_test_window(FILE *fp, int shift, double lambda, double *meanK_out, double *chi2_out, int
*df_out, double *p_out) { uint32_t bday[NO_BDAY]; uint32_t sp[NO_BDAY]; int obs[NO_OBS]; long
sumK = 0; for (int t = 0; t < NO_OBS; t++) { for (int i = 0; i < NO_BDAY; i++) { uint32_t x; if
(!read_u32_le(fp, &x)) { fprintf(stderr, "EOF: not enough data (shift=%d, sample=%d, i=%d)\n", shift,
t, i); exit(2); } bday[i] = (x >> shift) & MASK24; } qsort(bday, NO_BDAY, sizeof(uint32_t), cmp_u32); /*
spacings INCLUDING wrap-around spacing (critical) */ for (int i = 0; i < NO_BDAY - 1; i++) sp[i] = bday[i
+ 1] - bday[i]; sp[NO_BDAY - 1] = (bday[0] + N_DAYS) - bday[NO_BDAY - 1]; qsort(sp, NO_BDAY,
sizeof(uint32_t), cmp_u32); int K = 0; for (int i = 1; i < NO_BDAY; i++) if (sp[i] == sp[i - 1]) K++; obs[t] =
K; sumK += K; } *meanK_out = (double)sumK / (double)NO_OBS; *p_out = poisson_chisq_gof(obs,
NO_OBS, lambda, chi2_out, df_out); } int main(int argc, char **argv) { if (argc < 2) { fprintf(stderr,
"Usage: %s output.dat\n", argv[0]); return 1; } const char *filename = argv[1]; FILE *fp =
fopen(filename, "rb"); if (!fp) { perror("fopen"); return 1; } const double lambda = (double)NO_BDAY
* (double)NO_BDAY * (double)NO_BDAY / (4.0 * (double)N_DAYS); puts("\n\t|-----
-----|"); puts("\t| BIRTHDAY SPACINGS TEST (Clean C Version) |");

```

```
puts("\t|n=2^24, m=1024. For each 24-bit window, compute duplicates K |"); puts("\t|in spacing list
and test  $K \sim \text{Poisson}(\lambda=m^3/(4n)=16)$ . |"); puts("\t|Nine windows => 9 p-values => KS test for
Uniform(0,1). |"); puts("\t|-----| \n"); printf("\tRESULTS
FOR %s\n", filename); printf("\t(m=%d, n=2^%d, lambda=%.2f, samples=%d, windows=%d)\n\n",
NO_BDAY, NO_BITS, lambda, NO_OBS, WINDOWS); printf("\tBits used\tmean(K)\t\tchisq\t\ttdf\tp-
value\n"); double pvals[WINDOWS]; for (int shift = 0; shift < WINDOWS; shift++) { /* Diehard reuses
same file for each window: close/reopen or rewind */ rewind(fp); double meanK, chi2, p; int df;
birthday_test_window(fp, shift, lambda, &meanK, &chi2, &df, &p); int bit_lo = shift + 1; int bit_hi =
shift + NO_BITS; printf("\t%2d to %2d\t%.2f\t%.4f\t\t%d\t%.6g\n", bit_lo, bit_hi, meanK, chi2, df,
p); pvals[shift] = p; } fclose(fp); double ks_p = ks_uniform_pvalue(pvals, WINDOWS); printf("\n\tKS p-
value on %d p-values = %.6g\n\n", WINDOWS, ks_p); return 0; }
```

---

### What you must ensure about your outputdat

✓ It must be **binary** (raw bytes).

If you created it by printing numbers with `fprintf`, then it's **ASCII**, and this code won't work (it will read wrong bytes). If that's your case, tell me and I'll give an ASCII-reader version.

---

### One more critical note about your original code

Your original spacing loop:

```
for(i=no_bday-1; i>=1; --i) bdspace[i]=bdspace[i-1];
```

**misses the wrap-around spacing** unless you handle it elsewhere.

The clean code above fixes it by setting:

```
sp[NO_BDAY-1] = (bday[0] + N_DAYS) - bday[NO_BDAY-1];
```

If you want, paste **how you generate** outputdat (binary or text), and I'll match the reader exactly.

**You said:**

now explain mathematicalyy how bday test is ru ning and how chi sare test will appy

**ChatGPT said:**

#### 1) What data the Birthday Spacings test uses

You have a stream of 32-bit integers  $X_1, X_2, \dots$  from your file.

For each **bit-window** (there are 9 windows for 24 bits inside 32 bits), you extract a 24-bit "birthday"

$$Y_i = (X_i \gg s) \& (2^{24} - 1), s = 0, 1, \dots, 8.$$

So  $Y_i \in \{0, 1, \dots, N - 1\}$  with  $N = 2^{24}$ .

---

#### 2) One "sample" of the test (one $K$ )

Fix one window  $s$ . For one sample you take  $m = 1024$  birthdays:



$Y_1, \dots, Y_m$  i.i.d. uniform on  $\{0, \dots, N-1\}$ .

**Step A: sort**

$$Y_{(1)} \leq Y_{(2)} \leq \dots \leq Y_{(m)}.$$

**Step B: form the circular spacings** (this is important)

$$S_i = Y_{(i+1)} - Y_{(i)}, i = 1, \dots, m-1,$$

and the wrap-around spacing

$$S_m = (Y_{(1)} + N) - Y_{(m)}.$$

Now  $\sum_{i=1}^m S_i = N$ .

**Step C: count repeated spacings**

Let the multiset  $\{S_1, \dots, S_m\}$  be sorted, and define

$$K = \#\{i \in \{2, \dots, m\} : S_{(i)} = S_{(i-1)}\}.$$

So  $K$  counts how many “spacing collisions” you have (equal gaps).

That  $K$  is the **one number** produced by the test for that sample.

---

### 3) Why $K$ should be Poisson (the core math)

For large  $N$  and moderate  $m$ , the spacings behave like many “almost independent” draws from a large range. The event “two spacings are equal” is rare, and the total number of such rare collision events becomes approximately Poisson.

The classic asymptotic result used by Diehard is:

$$K \approx \text{Poisson}(\lambda), \lambda \approx \frac{m^3}{4N}.$$

For Diehard parameters  $m = 1024$  and  $N = 2^{24}$ :

$$\lambda = \frac{1024^3}{4 \cdot 2^{24}} = \frac{2^{30}}{2^{26}} = 2^4 = 16.$$

**Interpretation:** if your RNG is good, then each sample’s collision count  $K$  should look like a Poisson random variable with mean  $\approx 16$ . If your RNG has structure, you often get too many repeated spacings (or too few), so the distribution of  $K$  deviates from Poisson(16).

---

### 4) What Diehard actually does with many samples

For one window  $s$ , Diehard repeats the above **NO\_OBS = 500 times**, giving:

$$K_1, K_2, \dots, K_{500}.$$

Now it tests whether these 500 observed values look like they came from Poisson(16).

That's where **chi-square goodness-of-fit** comes in.

---

## 5) How the chi-square test is applied (mathematically)

### Step 1: build categories (bins) for $K$

Since  $K$  can be 0,1,2,3,..., you make bins like:

- Bin 1:  $K \in \{0,1,2, \dots, k_1\}$
- Bin 2:  $K \in \{k_1 + 1, \dots, k_2\}$
- ...
- Last bin:  $K \geq k_{r-1} + 1$

Diehard chooses bins **adaptively** so that each bin has **expected count  $\geq 5$**  (this is required so chi-square approximation is valid).

### Step 2: compute expected counts from Poisson( $\lambda$ )

For a bin  $B_j$ ,

$$E_j = n \cdot \Pr_{\text{Pois}(\lambda)}(K \in B_j), n = 500.$$

For a single value  $k$ :

$$\Pr(K = k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

For a range  $a \leq K \leq b$ :

$$\Pr(a \leq K \leq b) = \sum_{k=a}^b e^{-\lambda} \frac{\lambda^k}{k!}.$$

For a tail  $K \geq t$ :

$$\Pr(K \geq t) = 1 - \sum_{k=0}^{t-1} e^{-\lambda} \frac{\lambda^k}{k!}.$$

### Step 3: compute observed counts

From your 500 samples, count how many fall in each bin:

$$O_j = \#\{i: K_i \in B_j\}.$$

#### Step 4: chi-square statistic

$$\chi^2 = \sum_{j=1}^r \frac{(O_j - E_j)^2}{E_j}.$$

#### Step 5: degrees of freedom

Because  $\lambda$  is **not estimated from your data** (it is fixed at 16 from theory), the degrees of freedom are:

$$\nu = r - 1.$$

(If you had estimated  $\lambda$  from the sample mean, then it would be  $r - 2$ .)

#### Step 6: convert to p-value

Under the null hypothesis “ $K_i$  are Poisson(16)”:

$$\chi^2 \sim \chi_\nu^2.$$

So the p-value is the upper tail:

$$p = \Pr(\chi_\nu^2 \geq \chi_{\text{obs}}^2).$$

That p-value is what you print for that bit-window.

### 6) Why there is a KS test at the end

You have **9 windows**, so you get **9 p-values**:

$$p_1, \dots, p_9.$$

If the RNG is good, each p-value should be roughly Uniform(0,1), and collectively they should also look uniform.

So Diehard applies a **Kolmogorov–Smirnov test** to  $\{p_1, \dots, p_9\}$  against Uniform(0,1), producing one final KS p-value.

#### One-line summary (math)

- Extract 24-bit birthdays, sort, compute circular gaps, count repeated gaps  $K$ .
- Under randomness:  $K \approx \text{Poisson}(\lambda = m^3/(4N))$  with  $\lambda = 16$ .
- From 500 samples, do chi-square GOF of observed  $K$  histogram vs Poisson(16)  $\rightarrow$  p-value.

- Repeat for 9 bit windows → 9 p-values → KS test for uniformity.

If you want, I can write down an explicit small bin example (like how  $E_j$  is computed and merged until  $\geq 5$ ) using  $\lambda = 16$  and  $n = 500$ , so you can see exact numbers.

**You said:**

in easiest way please explain

**ChatGPT said:**

Okay — here's the easiest way to see it, with *exactly* what the test is doing.

**What Birthday Spacings is checking (one sentence)**

It checks whether your random numbers create **too many equal gaps** after sorting (equal gaps = "collisions").

**A) How the Birthday test runs (super simple)**

**Step 1: Take numbers from your file**

Your file is bytes → Diehard reads **4 bytes at a time** = one **32-bit integer**.

**Step 2: Make "birthdays" from it**

A "birthday" is just a number in a range of size  $2^{24}$  (0 to 16,777,215).

So from each 32-bit integer  $X$ , take 24 bits:

$$Y = (X \gg s) \& 0xFFFFFFFF$$

- $s$  is the shift (window)
- There are **9 windows**:  $s = 0, 1, \dots, 8$

**Step 3: For one trial, take  $m = 1024$  birthdays**

So you collect:

$$Y_1, Y_2, \dots, Y_{1024}$$

**Step 4: Sort them**

Example (tiny):

$$[2, 15, 7, 8] \Rightarrow [2, 7, 8, 15]$$

**Step 5: Compute the gaps between neighbors (spacings)**

For sorted birthdays:

$$S_1 = Y_2 - Y_1, S_2 = Y_3 - Y_2, \dots$$

Plus **wrap-around gap** (important):

$$S_m = (Y_1 + N) - Y_m, N = 2^{24}$$

So you get **1024 gap values**.

#### Step 6: Count repeated gaps

If the gap list contains equal values like:

$$[3, 1, 7, 1, 9, \dots]$$

then gap “1” repeats → that increases the count  $K$ .

So for that one trial you output:

- $K$  = number of repeated spacings

☑ That's one Birthday test trial.

---

#### B) Why Poisson (what Diehard expects)

If numbers are really random, repeated gaps are **rare events**.

Rare events counted many times usually follow **Poisson** distribution.

Diehard says:

$$K \approx \text{Poisson}(\lambda)$$

where:

$$\lambda = \frac{m^3}{4N}$$

Here:

- $m = 1024$
- $N = 2^{24}$

So:

$$\lambda = 16$$

Meaning: a good RNG should produce  $K$  values centered around ~16.

---

#### C) Where chi-square comes in (easiest explanation)

Diehard repeats the above trial **500 times** (for the same bit-window), so you get:

$$K_1, K_2, \dots, K_{500}$$

Now you ask:

**“Do these 500 numbers look like they come from Poisson(16)?”**

That is exactly what the **chi-square goodness-of-fit** test checks.

---

### Chi-square in plain steps

#### Step 1: Make a frequency table (histogram)

Count how many times you got:

- $K = 0$
- $K = 1$
- $K = 2$
- ...  
(and for large  $K$ , group them together)

These are **observed counts**:

$$O_0, O_1, O_2, \dots$$

#### Step 2: Compute expected counts from Poisson(16)

Poisson probability:

$$P(K = k) = e^{-16} \frac{16^k}{k!}$$

Expected count in 500 trials:

$$E_k = 500 \cdot P(K = k)$$

So:

- If Poisson says “ $K = 14$ ” should happen  $\sim 40$  times, then  $E_{14} \approx 40$ .
- Your data might give  $O_{14} = 55$ , etc.

#### Step 3: Compute chi-square number

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- If observed  $O$  is close to expected  $E$ , chi-square is small  $\rightarrow$  good.
- If very different, chi-square is large  $\rightarrow$  bad.

#### Step 4: Convert chi-square to a p-value

This p-value tells:

- **near 0** → your data does **not** match Poisson(16) → suspicious RNG.
- **middle (like 0.1 to 0.9)** → looks fine.

That p-value is the line Diehard prints for that window.

---

#### D) Why 9 windows + KS test

Diehard repeats everything for **9 different 24-bit slices** of the 32-bit words.

So you get **9 p-values**:

$$p_1, \dots, p_9$$

If RNG is good, p-values should look **uniform in (0,1)**.

So Diehard runs **KS test** on those 9 p-values (final check).

---

#### One super-short summary

1. Make 1024 “birthdays” from your bits
  2. Sort, take gaps, count repeated gaps →  $K$
  3. Repeat 500 times → 500  $K$ 's
  4. Check those 500  $K$ 's match Poisson(16) using **chi-square**
  5. Do for 9 bit windows, then KS test on 9 p-values
- 

If you want, I can show a **tiny numeric chi-square example** with fake data: like observed counts for  $K = 10..22$ , compute expected counts from Poisson(16), then compute  $\chi^2$  by hand so it becomes crystal clear.

**You said:**

like we have 1024 ,32 bit integer and then we will run it for a window and then find dife so we will get total 1023 diff na

**ChatGPT said:**

Almost — you'll get **1024 spacings**, not 1023.

Here's why, in the simplest way.

**You start with 1024 birthdays (from 1024 integers)**

From each 32-bit integer you take a 24-bit “birthday”:

$$Y_i = (X_i \gg s) \& 0xFFFFF$$

So you have:

$$Y_1, \dots, Y_{1024}$$

**After sorting, normal differences give 1023 gaps**

Sort:

$$Y_{(1)} \leq Y_{(2)} \leq \dots \leq Y_{(1024)}$$

Then consecutive differences:

$$D_i = Y_{(i+1)} - Y_{(i)}, i = 1, \dots, 1023$$

✓ Yes — **that part gives 1023 differences.**

**But Diehard adds one more “wrap-around” gap**

Because birthdays live on a circle of size  $N = 2^{24}$  (like days in a year), you also include the gap from the last back to the first:

$$D_{1024} = (Y_{(1)} + N) - Y_{(1024)}$$

So total spacings:

- 1023 consecutive gaps
- - 1 wrap-around gap  
= **1024 spacings**

**Why add wrap-around?**

So that all gaps together cover the whole “year”:

$$\sum_{i=1}^{1024} D_i = N.$$

Without wrap-around, you miss the end segment between the largest and smallest birthdays.

---

**Then what does the test count?**

You now have **1024 spacing values**.

You sort these spacings and count how many are repeated.

That repeat-count is the  $K$  for that one run.

---

✓ So your sentence becomes:



“We take 1024 32-bit integers → extract 24-bit birthdays for a window → sort → compute 1023 consecutive gaps **plus 1 wrap-around gap** → total 1024 spacings → count repeated spacings.”

If you want, I can show a 5-number toy example where you see exactly why wrap-around makes the count become  $m$  spacings instead of  $m - 1$ .

**You said:**

now we have 1024 spacing then wht to do please eaplin in simplest

**ChatGPT said:**

Sure — once you have **1024 spacings**, the test does only **one simple thing**:

**After you get the 1024 spacings, do this:**

**Step 1: Sort the spacings**

Example (small):

Spacings = [5, 2, 2, 9, 1, 5]

Sorted spacings = [1, 2, 2, 5, 5, 9]

**Step 2: Count how many times a spacing repeats**

Walk through the sorted list:

- compare each element with the previous one
- if same → it's a duplicate → increase counter  $K$

In the example:

- 2 equals previous 2 → +1
  - 5 equals previous 5 → +1
- So  $K = 2$ .

☑ **That  $K$  is the output of the Birthday test for ONE run.**

---

**Then what?**

Diehard does this **500 times** (each time using a fresh set of 1024 birthdays from the file):

So you get:

$$K_1, K_2, \dots, K_{500}$$

---

**Now apply Chi-square (still simple)**

**Step 3: Make a frequency table of the  $K$ 's**

Count how many times you got:

- $K = 10$

- $K = 11$
- $K = 12$
- ...  
etc.

These are **Observed counts**  $O$ .

#### Step 4: Compare with what you *should* get (Poisson with mean 16)

For a good RNG,  $K$  should behave like:

$$K \sim \text{Poisson}(\lambda = 16)$$

So expected count for value  $k$  is:

$$E_k = 500 \cdot P(K = k), P(K = k) = e^{-16} \frac{16^k}{k!}$$

#### Step 5: Compute chi-square number

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- If  $O$  close to  $E \rightarrow$  chi-square small  $\rightarrow$  looks random
- If  $O$  far from  $E \rightarrow$  chi-square large  $\rightarrow$  not random

Then chi-square gives a **p-value**.

#### One line summary (after 1024 spacings)

**Sort spacings  $\rightarrow$  count repeated spacings =  $K \rightarrow$  repeat 500 times  $\rightarrow$  check the 500  $K$ 's match Poisson(16) using chi-square.**

That's it.

If you want, I can show an ultra-small "fake" run (like 10 trials instead of 500) and do the counting table and chi-square idea with actual numbers so it becomes fully clear.

**You said:**

here what is  $k$  i mean i dint get is  $k$  is a maimum scount of any frquncy or else

**ChatGPT said:**

**$K$  is NOT the maximum frequency.**

In the Birthday Spacings test,  **$K$  = the number of duplicated spacings** in that one run.

**Think of it like this**

You have **1024 spacing values** (gaps). Some gaps may repeat.

1. Sort the spacings.
2. Walk through them and count how many times you see the *same value again*.

That count is **K**.

---

### Small example

Spacings (say you got 8 spacings):

[3, 7, 3, 1, 9, 7, 7, 2]

Sort:

[1, 2, 3, 3, 7, 7, 7, 9]

Now compare neighbors:

- 3 equals previous 3 → **+1**
- 7 equals previous 7 → **+1**
- next 7 equals previous 7 → **+1**

So:

$$K = 3$$

### Notice what happened

- spacing 3 appeared **2 times** → contributed **1** to K
- spacing 7 appeared **3 times** → contributed **2** to K

So in general:

- if a spacing value appears **r times**, it contributes **(r - 1)** to K

Equivalently:

$$K = \sum_{\text{each spacing value}} (r_v - 1) \text{ for all } r_v \geq 2$$


---

### What K is NOT

- Not the maximum count of any spacing (that would be  $\max r_v$ )
- Not the number of *distinct* spacing values
- Not the total number of spacings (that's 1024)

---

### Why Diehard uses K

Because K measures **how many repeats** you got in gaps.

A good RNG gives repeats at the “right” rate (around Poisson with mean 16 for  $m=1024$ ,  $n=2^{24}$ ).

If you want, I can show how K changes in three cases:

- all spacings different  $\rightarrow K=0$
- one spacing appears twice  $\rightarrow K=1$
- one spacing appears 10 times  $\rightarrow K=9$

**You said:**

yes

**ChatGPT said:**

Cool — here are the three cases, super clear.

#### Case 1: All spacings different

Spacings (example):

[1, 4, 7, 2, 9, 3]

Sorted:

[1, 2, 3, 4, 7, 9]

No equal neighbors  $\rightarrow$  **no repeats**

$$K = 0$$

☑ Meaning: **0 duplicated spacings**

---

#### Case 2: One spacing appears twice

Spacings:

[1, 4, 7, 2, 4, 9]

Sorted:

[1, 2, 4, 4, 7, 9]

There is one duplicate pair “4,4”  $\rightarrow$  counts **1**

$$K = 1$$

✓ Meaning: one spacing value repeated once.

---

### Case 3: One spacing appears 10 times

Suppose spacing “5” appears 10 times and others are different.

Spacings (toy):

[5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 1, 2, 3]

Sorted (same idea):

[1, 2, 3, 5, 5, 5, 5, 5, 5, 5, 5, 5]

Now how many duplicates does “5” create?

If a value appears  $r$  times, it contributes:

$$r - 1$$

Here  $r = 10$ , so contribution:

$$10 - 1 = 9$$

So:

$$K = 9$$

✓ Meaning: you had 9 “extra occurrences” beyond the first.

---

### Quick rule to remember

If the spacing counts are  $r_1, r_2, \dots$  for each distinct spacing value, then:

$$K = \sum (r_i - 1) \text{ over all } r_i \geq 2$$

So:

- 2 occurrences → adds 1 to  $K$
  - 3 occurrences → adds 2 to  $K$
  - 10 occurrences → adds 9 to  $K$
- 

If you want, I can connect this to the next step: **how those  $K$ 's (500 of them) are compared to Poisson(16)** in one simple picture-style explanation.

**You said:**

now in one window i have find k now what to do expain in dimplr

**ChatGPT said:**

In **one window**, you don't stop after finding **one**  $K$ . You must do it **500 times**.

**What to do after you find  $K$  (simplest)**

**Step 1: Repeat the same process 500 times**

Each time:

- take **1024 new birthdays** from the file
- sort → make 1024 spacings → count duplicates
- you get one  $K$

So you will collect:

$$K_1, K_2, \dots, K_{500}$$

---

**Step 2: Check if these 500 values look like Poisson(16)**

Diehard expects:

$$K \sim \text{Poisson}(\lambda = 16)$$

So you compare your 500 observed  $K$ 's to what Poisson(16) predicts using **chi-square**.

---

**How chi-square is done (very simple)**

**(a) Make counts (observed)**

Count how many times you got each  $K$ :

- how many times  $K = 10$
- how many times  $K = 11$
- ...

These are **Observed** counts  $O$ .

**(b) Compute expected counts from Poisson(16)**

For each  $k$ :

$$E(k) = 500 \times P_{\text{Pois}(16)}(k)$$

(where  $P_{\text{Pois}(16)}(k) = e^{-16} \frac{16^k}{k!}$ )

**(c) Compute chi-square number**

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

**(d) Convert to a p-value**

That p-value is the result for this window:

- p-value very small (near 0)  $\Rightarrow$  bad (not matching Poisson)
  - p-value not extreme  $\Rightarrow$  ok
- 

**Step 3 (Diehard final): do this for all 9 windows**

You get 9 p-values (one per window).

Then apply **KS test** on those 9 p-values (they should look Uniform(0,1)).

---

**One-line summary**

**In one window:** compute  $K$  500 times  $\rightarrow$  chi-square vs Poisson(16)  $\rightarrow$  get 1 p-value.

If you want, I can show a tiny toy version with **10 trials instead of 500**, so you see exactly how the “count table  $O$ ” and “expected table  $E$ ” are formed.

**You said:**

for 500 round i have found  $k_1, k_2, \dots, k_{500}$  now what to do please explain in simple

**ChatGPT said:**

Great — now you have  $K_1, K_2, \dots, K_{500}$ .

Next you do only **one thing**:

**You check: “Do these 500 K-values behave like Poisson(16)?”**

That check is done with a **chi-square goodness-of-fit** test.

---

**What you do (step by step, simplest)**

**Step 1: Make a count table (Observed)**

Count how many times each  $K$  occurred.

Example idea (not real numbers):

- $K = 10$  occurred 28 times
- $K = 11$  occurred 35 times
- $K = 12$  occurred 41 times

- ...

These counts are called:

$$O_k = \#\{i: K_i = k\}$$

(For large K values, you group them together so counts aren't tiny.)

## Step 2: Compute what you *should* expect (Expected) from Poisson(16)

Poisson probability:

$$P(K = k) = e^{-16} \frac{16^k}{k!}$$

Expected count in 500 trials:

$$E_k = 500 \cdot P(K = k)$$

So if Poisson says  $P(K = 12) = 0.07$ , then:

$$E_{12} = 500 \times 0.07 = 35$$

## Step 3: Compute the chi-square value

For each bin/value:

$$\frac{(O-E)^2}{E}$$

Add them all:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- If observed  $O$  is close to expected  $E$ ,  $\chi^2$  is small  $\rightarrow$  good.
- If very different,  $\chi^2$  is big  $\rightarrow$  bad.

## Step 4: Convert $\chi^2$ to a p-value

Using the chi-square distribution (with proper degrees of freedom), you get a **p-value**.

That **p-value is your final result for this one window**.

- p-value near **0** (like 0.00001)  $\rightarrow$  fails (not Poisson-like)



- p-value near **1** repeatedly → also suspicious
- p-value around **0.1–0.9** → looks fine

### That's it for ONE window

Then you repeat the whole thing for the other windows to get 9 p-values and do the KS test.

If you want, paste 20–30 of your  $K_i$  values (or the frequency table you got), and I'll show you exactly how to form bins and compute  $O$ ,  $E$ , and  $\chi^2$  in the simplest way.

**You said:**

did not get how u r calculatg obserbed and expected value please explain in simple

**ChatGPT said:**

Let's make it **very concrete** with a tiny example. Same steps work for your 500 values — just bigger.

**You have 500 numbers  $K_1, \dots, K_{500}$**

Each  $K$  is “how many duplicated spacings” in that run.

#### **1) Observed values $O$ : just COUNT what you got**

Suppose (toy example) you ran only 20 times and got:

$$K = [14, 16, 15, 18, 16, 16, 12, 14, 17, 15, 16, 13, 14, 16, 19, 15, 16, 14, 15, 16]$$

Now count frequency of each value:

- How many 12? → 1
- How many 13? → 1
- How many 14? → 4
- How many 15? → 4
- How many 16? → 7
- How many 17? → 1
- How many 18? → 1
- How many 19? → 1

These counts are the **observed counts**:

$$O_{12} = 1, O_{13} = 1, O_{14} = 4, O_{15} = 4, O_{16} = 7, O_{17} = 1, O_{18} = 1, O_{19} = 1$$

That's it. **Observed** = “how many times each  $K$  happened.”

For your real case, you do the same with 500 values.

## 2) Expected values $E$ : what Poisson(16) predicts

Diehard expects:

$$K \sim \text{Poisson}(16)$$

Poisson tells probability of seeing exactly  $k$ :

$$P(K = k) = e^{-16} \frac{16^k}{k!}$$

Now since you did **500 trials**, expected count is:

$$E_k = 500 \times P(K = k)$$

**Example:**

If Poisson says  $P(K = 16) = 0.10$  (just example), then:

$$E_{16} = 500 \times 0.10 = 50$$

Meaning: **you expect about 50 times** to see  $K = 16$  in 500 trials.

---

## 3) But there is one practical rule

Chi-square needs each expected bin count to be about  $\geq 5$ .

So you **group** rare values:

Example bins (common style):

- Bin 1:  $K \leq 10$
- Bin 2:  $K = 11$
- Bin 3:  $K = 12$
- ...
- Bin last:  $K \geq 25$

**Observed in a bin:**

Add the counts inside that bin.

Example:

If bin is  $K \leq 10$ :

$$O_{\leq 10} = O_0 + O_1 + \dots + O_{10}$$

**Expected in a bin:**

Add probabilities inside that bin then multiply by 500:

$$E_{\leq 10} = 500 \times \sum_{k=0}^{10} P(K = k)$$

For a tail bin  $K \geq 25$ :

$$E_{\geq 25} = 500 \times \left( 1 - \sum_{k=0}^{24} P(K = k) \right)$$

## Super-simple summary

### Observed $O$

👉 **Count from your data.**

“how many times did each  $K$  (or each bin) happen?”

### Expected $E$

👉 **Poisson prediction.**

“probability from  $\text{Poisson}(16) \times 500$  trials”

If you paste just a **frequency table** from your 500 values (like: “ $K=10$  occurred 8 times,  $K=11$  occurred 14 times, ...”), I will compute the **expected counts** and show exactly how the bins are grouped and how chi-square is formed, step-by-step.

### You said:

how we find expected value

### ChatGPT said:

To find the **expected value (expected count)**, you do this:

#### 1) Expected count for one exact value $k$

Diehard assumes:

$$K \sim \text{Poisson}(\lambda = 16)$$

So probability of getting exactly  $k$  is:

$$P(K = k) = e^{-16} \frac{16^k}{k!}$$

If you ran the experiment  $n = 500$  times, then the **expected number of times** you should see  $k$  is:

$$E_k = n \times P(K = k) = 500 \times e^{-16} \frac{16^k}{k!}$$

That's the expected value (expected **count**) for  $k$ .

---

## 2) Expected count for a BIN (range), like $K \leq 10$ or $12 \leq K \leq 15$

You add probabilities in that range:

**Example: expected count for  $12 \leq K \leq 15$**

$$E_{12..15} = 500 \times \sum_{k=12}^{15} e^{-16} \frac{16^k}{k!}$$

**Example: expected count for tail  $K \geq 25$**

$$E_{\geq 25} = 500 \times \left( 1 - \sum_{k=0}^{24} e^{-16} \frac{16^k}{k!} \right)$$

---

## 3) What "expected value" means in simple words

- Poisson tells you "how likely each  $k$  is"
  - Multiply by how many trials you did (500)
  - That gives "how many times you *expect* to see it"
- 

## Quick numeric example (just to feel it)

Suppose Poisson(16) gives  $P(K = 16) = 0.10$  (example).

Then:

$$E_{16} = 500 \times 0.10 = 50$$

Meaning: you expect around **50** occurrences of  $K = 16$  in 500 trials.

---

If you tell me one value you care about (say  $k = 16$  or  $k = 10$ ), I can compute the actual  $P(K = k)$  and the expected count  $E_k$  numerically.

**You said:**

How many 12? → 1 like here observed  $o_i=1$  so what is expected for this

**ChatGPT said:**

Thought for 7s

In that example you had **Observed** for  $k = 12$ :

$$O_{12} = 1$$

Now **Expected** means: if  $K \sim \text{Poisson}(16)$ , how many times should  $K = 12$  appear in 500 trials?

**Step 1: Poisson probability for  $k = 12$**

$$P(K = 12) = e^{-16} \frac{16^{12}}{12!} \approx 0.06613$$

**Step 2: Expected count in 500 trials**

$$E_{12} = 500 \times P(K = 12) \approx 500 \times 0.06613 \approx 33.06$$

✓ So for  $k = 12$ :

- **Observed:**  $O_{12} = 1$
- **Expected:**  $E_{12} \approx 33.06$

That big difference would contribute a lot to chi-square:

$$\frac{(O_{12} - E_{12})^2}{E_{12}} = \frac{(1 - 33.06)^2}{33.06} \approx 31.1$$

(And in real Diehard, you usually **bin/merge** values so each expected bin is  $\geq 5$ ; but the *expected for exactly 12* is about **33**.)

**You said:**

how in diehard it will work explain with example and find then chi square observed and expected and i know already how to find  $k_1, k_2, \dots, k_{500}$

**ChatGPT said:**

Thought for 1m 18s

Yes — you already have  $K_1, \dots, K_{500}$  for **one window**.

Now Diehard does **only this**:

**What Diehard does next (1 window)**

**Step 1: Make Observed counts (just counting)**

Take your 500 values and count how many times each  $K$  happens.

Example: if among 500 runs you got

- $K = 12$  occurs 33 times  $\rightarrow$  **Observed**  $O_{12} = 33$
- $K = 16$  occurs 49 times  $\rightarrow$  **Observed**  $O_{16} = 49$

That's all "Observed" means: **how many you actually saw**.

---

**Step 2: Make Expected counts (from Poisson(16))**

For Birthday Spacings with  $m = 1024$ ,  $N = 2^{24}$ ,

$$\lambda = \frac{m^3}{4N} = 16.$$

So Diehard assumes:

$$K \sim \text{Poisson}(16).$$

Poisson probability:

$$P(K = k) = e^{-16} \frac{16^k}{k!}.$$

Since you have 500 runs, the **expected number** of times you should see  $k$  is:

$$E_k = 500 \times P(K = k).$$

Example (for  $k = 12$ ):

$$P(K = 12) \approx 0.06613 \Rightarrow E_{12} \approx 500 \times 0.06613 \approx 33.06.$$

So “Expected” means: **Poisson says you should see it about  $E_k$  times.**

### Step 3: (Important) Combine rare values into bins

Chi-square needs expected counts not too small (rule: **each expected bin  $\geq 5$** ).

So Diehard groups like this (one valid simple choice):

- Bin A:  $K = 0, 1, 2, \dots, 8$  (all together)
- Then single bins:  $K = 9, 10, 11, \dots, 24$
- Bin last:  $K \geq 25$

Now you compute:

- **Observed per bin** = how many of your 500  $K$ 's fall in that bin
- **Expected per bin** =  $500 \times$  (Poisson probability of that bin)

Example:

$$E_{0-8} = 500 \sum_{k=0}^8 P(K = k), E_{25+} = 500 \left( 1 - \sum_{k=0}^{24} P(K = k) \right).$$

### Step 4: Compute chi-square

For each bin  $j$ :

$$\chi^2 = \sum_j \frac{(O_j - E_j)^2}{E_j}.$$

Degrees of freedom:

$$df = (\text{\#bins}) - 1$$

(because  $\lambda = 16$  is fixed, not estimated).

Then convert  $\chi^2$  to a **p-value** using the chi-square distribution.

That p-value is the **result for this window**.

### Full worked example (with numbers)

Suppose from your 500 runs (one window), your **Observed** counts were:

| Bin | Observed $O$ | Expected $E$ (Poisson(16) $\times$ 500) |
|-----|--------------|-----------------------------------------|
| 0–8 | 11           | 10.99                                   |
| 9   | 17           | 17.05                                   |
| 10  | 26           | 27.28                                   |
| 11  | 43           | 39.69                                   |
| 12  | 30           | 52.92                                   |
| 13  | 55           | 65.14                                   |
| 14  | 62           | 74.45                                   |
| 15  | 55           | 79.41                                   |
| 16  | 49           | 79.41                                   |
| 17  | 59           | 74.74                                   |

| Bin | Observed $O$ | Expected $E$ (Poisson(16) $\times$ 500) |
|-----|--------------|-----------------------------------------|
| 18  | 47           | 66.44                                   |
| 19  | 35           | 55.95                                   |
| 20  | 26           | 44.76                                   |
| 21  | 26           | 34.10                                   |
| 22  | 19           | 24.80                                   |
| 23  | 16           | 17.25                                   |
| 24  | 7            | 11.50                                   |
| 25+ | 17           | 11.16                                   |

Now compute chi-square:

$$\chi^2 = \sum \frac{(O-E)^2}{E}.$$

For one row example (say  $k = 12$ ):

$$\frac{(30-52.92)^2}{52.92} \approx 9.93.$$

Do this for all rows and add  $\rightarrow$  here it gives:

- $\chi^2 \approx 16.34$
- number of bins = 18  $\Rightarrow df = 17$
- chi-square p-value  $\approx 0.50$  (this looks fine).

#### Final picture (one window)

1. You already have  $K_1, \dots, K_{500}$ .
2. Count them into bins  $\rightarrow$  that's **Observed  $O$** .
3. Compute Poisson(16) probabilities  $\rightarrow$  multiply by 500  $\rightarrow$  **Expected  $E$** .



4. Compute  $\chi^2 = \sum (O - E)^2 / E$ .

5. Convert  $\chi^2$  to a p-value.

If you paste your **frequency table** (e.g., "K=0 occurs ..., K=1 occurs ..." or your binned counts), I'll compute the exact  $E$ ,  $\chi^2$ , and p-value for your real data.

**You said:**

one doubt in original diehard how many bins used

**ChatGPT said:**

In the **original Diehard** code (like the one you pasted), the number of bins is **not fixed**.

**What it does**

- It sets

$$\text{dim} = \frac{\text{no\_obs}}{5}$$

So with no\_obs = 500, it allocates space for up to:

$$\text{dim} = 100$$

(this is just a safe maximum)

- Then it **builds bins dynamically**:

It starts with  $i = -1$ . For each bin  $j$ :

- It keeps adding Poisson probabilities  $P(K = i)$  until the **expected count** in that bin reaches at least 5:

$$E_j = 500 \sum_{k \in \text{bin}} P(K = k) \geq 5$$

- That forms **one bin**.
- Repeat for the next bin, continuing with the next  $i$ .
- At the end, when the **remaining expected mass** (rest) would be  $< 5$ , it merges everything into the final bin.

**So how many bins does it end up using?**

It ends up with some number  $r$  such that:

- each bin has expected  $\geq 5$
  - total expected = 500
- So roughly:

$$r \approx \frac{500}{5} = 100$$

But in practice it's **much less than 100** for Poisson(16), because many single  $k$ 's already have expected count  $> 5$ .

✓ **Answer:** original Diehard uses a **variable number of bins**, typically **around 20–30 bins** for Poisson( $\lambda = 16$ ) with 500 samples (not exactly fixed), and the code stores the final count in `dgf` (degrees of freedom), where:

$$df = r - 1.$$

### How to know the exact number for your run

In your code, after `P_fit`, it prints:

```
printf("degree of freedoms is: %d\n", dgf);
```

So the number of bins used is:

$$r = dgf + 1.$$

If you want, tell me what `dgf` prints in your run, and I'll tell you exactly how many bins it created.

**You said:**

the code u give in that how much bins used

**ChatGPT said:**

In the **clean code I gave you**, the number of bins is also **NOT fixed** (same spirit as original Diehard).

### How many bins it uses

It builds bins dynamically so that **each bin has expected count  $\geq 5$** .

- You have `NO_OBS = 500`
- Minimum expected per bin = 5

So the **maximum possible** bins is:

$$500/5 = 100$$

But in practice (for Poisson(16)) it will be **much smaller**, typically around **~20–30 bins**, because many single  $k$  values already have expected  $> 5$ .

### Exact answer for your run

The code prints `df` (degrees of freedom).

Number of bins used is:

$$\text{bins} = df + 1.$$

So when you run it, look at the printed `df`.

Example: if it prints `df = 23`, then bins used = **24**.

If you paste what df prints for one window, I'll tell you exactly how many bins it used there.

**You said:**

```
/* Clean Diehard BIRTHDAY SPACINGS test Parameters (classic Diehard): n = 2^24 days m = 1024
birthdays per sample no_obs = 500 samples per bit-window windows = 9 (24-bit windows across 32-
bit words) For each window: - compute K (duplicate spacings) for each sample - do chi-square GOF to
Poisson(lambda = m^3/(4n) = 16) - output p-value Then: - KS test on the 9 p-values Build: gcc -O2 -
std=c11 bday_clean.c -lm Run: ./a.out output.dat */ #include <stdio.h> #include <stdlib.h> #include
<stdint.h> #include <string.h> #include <math.h> #define NO_OBS 500 #define NO_BDAY 1024
#define NO_BITS 24 #define N_DAYS (1u << 24) #define MASK24 0xFFFFFu #define WINDOWS (32 -
NO_BITS + 1) /* 9 */ static int cmp_u32(const void *a, const void *b) { uint32_t x = *(const
uint32_t*)a, y = *(const uint32_t*)b; return (x > y) - (x < y); } static int cmp_d(const void *a, const
void *b) { double x = *(const double*)a, y = *(const double*)b; return (x > y) - (x < y); } static int
cmp_i(const void *a, const void *b) { int x = *(const int*)a, y = *(const int*)b; return (x > y) - (x < y); }
/* ----- read one little-endian uint32 from raw byte file ----- */ static int read_u32_le(FILE *fp,
uint32_t *out) { unsigned char b[4]; if (fread(b, 1, 4, fp) != 4) return 0; *out = (uint32_t)b[0] |
((uint32_t)b[1] << 8) | ((uint32_t)b[2] << 16) | ((uint32_t)b[3] << 24); return 1; } /* ----- Regularized
upper incomplete gamma Q(a,x) for chi-square SF ----- */ /* (Numerical Recipes style
implementation; stable and standard.) */ static double gammln(double xx) { static const double
cof[6] = { 76.18009172947146, -86.50532032941677, 24.01409824083091, -1.231739572450155,
0.1208650973866179e-2, -0.5395239384953e-5 }; double x = xx, y = xx; double tmp = x + 5.5; tmp -=
(x + 0.5) * log(tmp); double ser = 1.000000000190015; for (int j = 0; j < 6; j++) ser += cof[j] / ++y;
return -tmp + log(2.5066282746310005 * ser / x); } static void gser(double *gamser, double a, double
x, double *gln) { const int ITMAX = 200; const double EPS = 3.0e-14; *gln = gammln(a); if (x <= 0.0) {
*gamser = 0.0; return; } double sum = 1.0 / a, del = sum, ap = a; for (int n = 1; n <= ITMAX; n++) { ap
+= 1.0; del *= x / ap; sum += del; if (fabs(del) < fabs(sum) * EPS) break; } *gamser = sum * exp(-x + a *
log(x) - (*gln)); } static void gcf(double *gammcf, double a, double x, double *gln) { const int ITMAX =
200; const double EPS = 3.0e-14; const double FPMIN = 1.0e-300; *gln = gammln(a); double b = x +
1.0 - a; double c = 1.0 / FPMIN; double d = 1.0 / b; double h = d; for (int i = 1; i <= ITMAX; i++) {
double an = -i * (i - a); b += 2.0; d = an * d + b; if (fabs(d) < FPMIN) d = FPMIN; c = b + an / c; if (fabs(c)
< FPMIN) c = FPMIN; d = 1.0 / d; double del = d * c; h *= del; if (fabs(del - 1.0) < EPS) break; }
*gammcf = exp(-x + a * log(x) - (*gln)) * h; } static double gammq(double a, double x) { if (x < 0.0 || a
<= 0.0) return 0.0; double gamser, gammcf, gln; if (x < a + 1.0) { gser(&gamser, a, x, &gln); return 1.0 -
gamser; } else { gcf(&gammcf, a, x, &gln); return gammcf; } } static double chisq_sf(int df, double
chi2) { return gammq(0.5 * df, 0.5 * chi2); } /* ----- Poisson pmf by recurrence: p(k) = e^-lambda lambda^k / k! -
----- */ static double poisson_pmf(int k, double lambda) { double p = exp(-lambda); for (int i = 1; i <=
k; i++) p *= lambda / (double)i; return p; } /* ----- Chi-square GOF to Poisson(lambda), bins
combined so expected >= 5 ----- Returns p-value, and outputs chi2 and df. */ static double
poisson_chisq_gof(const int *obs, int n, double lambda, double *chi2_out, int *df_out) { int *x =
(int*)malloc((size_t)n * sizeof(int)); if (!x) { fprintf(stderr, "malloc failed\n"); exit(1); } memcpy(x, obs,
(size_t)n * sizeof(int)); qsort(x, (size_t)n, sizeof(int), cmp_i); double chi2 = 0.0; int bins = 0; int idx = 0;
int k = -1; while (idx < n) { double E = 0.0; int O = 0; /* grow a bin until E >= 5 */ while (E < 5.0) { k++; E
+= (double)n * poisson_pmf(k, lambda); while (idx < n && x[idx] <= k) { O++; idx++; } /* if we already
consumed all obs, stop */ if (idx >= n) break; } if (E > 0.0) { double d = (double)O - E; chi2 += (d * d) /
E; bins++; } } free(x); int df = bins - 1; /* lambda fixed (not estimated) */ if (df < 1) df = 1; *chi2_out =
chi2; *df_out = df; return chisq_sf(df, chi2); } /* ----- 1-sample KS test against Uniform(0,1),
asymptotic p-value ----- */ static double ks_uniform_pvalue(const double *pvals, int n) { double *x
= (double*)malloc((size_t)n * sizeof(double)); if (!x) { fprintf(stderr, "malloc failed\n"); exit(1); }
```

✓ **Answer:** For each window, **bins used = (printed df) + 1**.

---

### Why it varies

The code builds each bin by grouping consecutive  $k$  values until the **expected count** in that bin reaches  $\geq 5$ :

```
while (E < 5.0) { k++; E += n * poisson_pmf(k, lambda); ... } bins++;
```

So if one  $k$  already has expected  $\geq 5$ , that  $k$  becomes a bin alone.

If expected is small, several  $k$ 's are merged into one bin.

---

### Typical value (for Poisson(16), $n=500$ )

For  $\lambda = 16$  and  $n = 500$ , most middle  $k$  values have large expected counts, so the bins end up roughly:

- a few merged bins in the far-left tail,
- many single- $k$  bins around the peak,
- a few merged bins in the far-right tail.

So usually you'll see **about 20–30 bins**, but the **exact** is whatever `df+1` prints.

---

### If you want the code to print bins directly

Add this print after `birthday_test_window(...)` call in main:

```
printf("\t(bins used = %d)\n", df + 1);
```

That will show it for each window.

**You said:**

BUT how we will know `df`

**ChatGPT said:**

In *your code*, you **already compute** `df` here in main:

```
double meanK, chi2, p; int df; birthday_test_window(fp, shift, lambda, &meanK, &chi2, &df, &p);
```

So **you know** `df` because it's the integer variable `df` that gets filled by `birthday_test_window()` (which gets it from `poisson_chisq_gof()`).

### 1) Easiest: just print it (you already do)

You already print `df` in this line:

```
printf("\t%2d to %2d\t%.2f\t%.4f\t%.6g\n", bit_lo, bit_hi, meanK, chi2, df, p);
```

So when you run the program, you'll see output like:

```
1 to 24  15.98  21.34  23  0.56
```

Here:

- $df = 23$
- $\text{bins used} = df + 1 = 24$

---

## 2) If you want it even clearer: print bins too

Modify the print line to include bins:

```
printf("\t%2d to %2d\t%.2f\t%.4f\t%.6g\t(bins=%d)\n", bit_lo, bit_hi, meanK, chi2, df, p, df+1);
```

Now output will directly show:

```
1 to 24  15.98  21.34  23  0.56  (bins=24)
```

---

## 3) Where df comes from (simple)

Inside `poisson_chisq_gof()`:

- it counts how many bins it created  $\rightarrow$  bins
- then sets:

```
df = bins - 1;
```

So **df is not something you guess** — it's computed and printed automatically.

If you run your program and paste one output line, I can tell you immediately how many bins were used in that window.

**You said:**

how df is computed please explain i don't know

**ChatGPT said:**

Sure. **df** means **degrees of freedom**. In this chi-square test it is basically:

**$df = (\text{number of bins}) - 1$**

That's it.

Now I'll explain **how the code decides the number of bins** and then how it turns that into **df**.

---

## 1) Why do we need "bins" at all?

You have 500 values  $K_1, \dots, K_{500}$ .

We want to compare them to **Poisson(16)**.

To do chi-square, we don't compare each individual  $K$  directly.

We first group them into **bins** like:

- Bin 1:  $K = 0,1,2$
- Bin 2:  $K = 3,4$
- Bin 3:  $K = 5$
- Bin 4:  $K = 6$
- ...

But we must ensure:

✓ **Each bin's expected count is at least 5**  
(because chi-square approximation works well only then)

---

## 2) How the code builds bins (very simple)

Inside `poisson_chisq_gof()` the code does:

**Start with  $k = -1$ , bins=0**

`int bins = 0; int k = -1;`

**Create one bin like this:**

- set  $E=0$  (expected in this bin)
- keep adding Poisson probabilities for  $k=0,1,2,\dots$  until  $\text{expected} \geq 5$ :

`while (E < 5.0) { k++; E += n * poisson_pmf(k, lambda); } bins++;`

Where:

- $n = 500$
- $\text{lambda} = 16$
- $\text{poisson\_pmf}(k, 16) = P(K=k)$
- so  $n * \text{poisson\_pmf}(\dots) = \text{expected count for that } k$

So one bin may contain:

- only one value (like only  $k=15$ ), if its expected  $\geq 5$
  - or several values (like  $k=0..7$  together), if each is tiny
- 

## 3) Example of bin building (intuitive)

For `Poisson(16)`:

- $P(K = 0)$  is extremely small  $\rightarrow$  expected in 500 trials maybe 0.0000...
- $P(K = 1)$  also tiny
- so the code keeps adding:

- expected for  $k=0$
- - expected for  $k=1$
- - expected for  $k=2$
- ...
  - until the sum becomes  $\geq 5$

That becomes **bin #1**.

Then it starts the next bin at the next  $k$  values, and repeats.

So it might end up with bins like:

- Bin 1:  $k = 0..8$
- Bin 2:  $k = 9$
- Bin 3:  $k = 10$
- ...
- Last bin:  $k = 26..\infty$  (tail)

The exact split depends on the expected counts.

---

#### 4) Now how $df$ is computed

After the loop ends, you have:

$bins = (\text{some number})$   $df = bins - 1$ ;

So:

$$df = bins - 1$$

#### Why “minus 1”?

Because once you know counts in all bins, the last one is automatically fixed:

- Total observed counts must sum to 500
- Total expected counts also sum to 500

So only  $(bins-1)$  counts are “free”.

---

#### 5) How many bins? (rough idea)

Since each bin has expected  $\geq 5$ , and total expected = 500:



$$\text{bins} \leq \frac{500}{5} = 100$$

In practice for Poisson(16) it's usually around **20–30**, so df around **19–29**.

### One-line takeaway

- ✓ The code keeps making bins until each bin has **expected  $\geq 5$** .
- ✓ It counts how many bins it made = bins.
- ✓ Then it sets:

$$\boxed{\text{df} = \text{bins} - 1}$$

If you want, I can show you how to print the actual bin ranges (like “bin1: 0–8, bin2: 9, ...”) from the code, so you can literally see the bins.

**You said:**

first what is calculate bind and df which one

**ChatGPT said:**

You calculate **bins first**, then you get **df** from bins.

### 1) What are “bins”?

Bins are just **groups of K-values** you put together for the chi-square test.

Example bins might look like:

- Bin 1:  $K = 0$  to 8
- Bin 2:  $K = 9$
- Bin 3:  $K = 10$
- ...
- Last bin:  $K \geq 25$

Diehard makes bins so that **each bin has expected count  $\geq 5$** .

### 2) What is df?

df = **degrees of freedom** for the chi-square distribution.

Once you know how many bins you ended up with:

$$\boxed{\text{df} = \text{(#bins)} - 1}$$

**Order (which comes first)**

- ✓ **First:** create bins
- ✓ **Then:** compute df from bins

That's exactly what your code does:

```
bins++; // each time a bin is created df = bins - 1; // after bins are done
```

**You said:**

how ill i know how many bins should create

**ChatGPT said:**

You **don't choose a fixed number of bins** in Diehard.

Diehard decides the number of bins **automatically** using one simple rule:

**Keep adding K-values into the current bin until the expected count in that bin is at least 5.**

Then close that bin and start the next one.

So you "know" how many bins only **after** you apply this rule from left tail to right tail.

---

### The rule in the simplest form

You have  $n = 500$  samples and  $\lambda = 16$ .

For each possible  $k = 0, 1, 2, \dots$ :

1. Compute expected count for that  $k$ :

$$E(k) = 500 \times P_{\text{Pois}(16)}(k)$$

2. Add it into the current bin's expected total  $E_{\text{bin}}$ .
3. When  $E_{\text{bin}} \geq 5$ , **finish this bin** and start a new bin.

That's exactly what your code does with:

```
while (E < 5.0) { k++; E += n * poisson_pmf(k, lambda); } bins++;
```

---

### How many bins should you create?

**As many as needed until all 500 samples are covered**, with the "expected  $\geq 5$ " rule.

#### Practical bounds

- Minimum expected per bin = 5
- Total expected = 500

So maximum bins is:

$$500/5 = 100$$

But for  $\text{Poisson}(16)$ , you typically get **~20–30 bins** (because near the center each single  $k$  already has expected  $> 5$ ).

---

### How YOU will know the number (in your run)

Your program prints df.

Then:

$$\text{bins} = df + 1$$

So just run it and read the line:

... df ...

Example:

- if it prints df = 23, then bins = 24.

---

### If you want it to literally show bin ranges (best for understanding)

Tell me and I'll modify your code to print:

- "Bin 1: k=0..8, Expected=5.3, Observed=6"
- "Bin 2: k=9..9, Expected=17.0, Observed=15"
- etc.

That will make the whole bin/df thing totally obvious.

**You said:**

so explain by taking example and give table

**ChatGPT said:**

Thought for 1m 29s

Sure — let's do **exactly what Diehard does (one window)** with a **real worked example table**.

We assume (as Diehard does for Birthday Spacings):

- you already computed  $K_1, \dots, K_{500}$
- theory says  $K \sim \text{Poisson}(\lambda = 16)$

---

### Step 1: What "expected" means

For any value  $k$ ,

$$P(K = k) = e^{-16} \frac{16^k}{k!}$$

and because you have **500 runs**,

$$E(k) = 500 \times P(K = k)$$

Example:

- $E(12) = 500 \times P(K = 12) \approx 500 \times 0.06613 \approx 33.06$

So **Expected** is “Poisson predicts you should see  $k$  about  $E(k)$  times in 500 runs.”

---

### Step 2: How Diehard chooses bins (most important)

Chi-square needs each bin’s expected count  $\geq 5$ , so Diehard groups small-probability  $k$ ’s.

**Rule:** starting from  $k = 0$ , keep adding  $k$ ’s into the current bin until the bin’s expected count reaches **at least 5**, then start the next bin.

For Poisson(16) with 500 runs, the bins come out like this:

- Bin1:  $K = 0$  to 8 (all together)
- then single bins: 9,10,11, ...,24
- last bin:  $K \geq 25$

That is **18 bins total**, so:

$$df = 18 - 1 = 17$$

---

### Step 3: “Observed” and “Expected” with a full example table

Below is an example (think of it as what you’d get after counting your 500 values into those bins).

- **Observed  $O$**  = how many of your 500  $K_i$ ’s fell in that bin
- **Expected  $E$**  = what Poisson(16) predicts for that bin
- Last column is the chi-square contribution  $(O-E)^2/E$

| Bin (K) | Observed O | Expected E (Poisson16×500) | (O-E) <sup>2</sup> /E |
|---------|------------|----------------------------|-----------------------|
| 0–8     | 9          | 10.99                      | 0.362                 |
| 9       | 6          | 10.66                      | 2.034                 |
| 10      | 21         | 17.05                      | 0.916                 |
| 11      | 23         | 24.80                      | 0.130                 |
| 12      | 32         | 33.06                      | 0.034                 |

| Bin (K) | Observed O | Expected E (Poisson16×500) | (O−E) <sup>2</sup> /E |
|---------|------------|----------------------------|-----------------------|
| 13      | 50         | 40.69                      | 2.128                 |
| 14      | 43         | 46.51                      | 0.265                 |
| 15      | 54         | 49.61                      | 0.389                 |
| 16      | 48         | 49.61                      | 0.052                 |
| 17      | 48         | 46.69                      | 0.037                 |
| 18      | 41         | 41.50                      | 0.006                 |
| 19      | 37         | 34.95                      | 0.120                 |
| 20      | 25         | 27.96                      | 0.313                 |
| 21      | 19         | 21.30                      | 0.249                 |
| 22      | 12         | 15.49                      | 0.787                 |
| 23      | 9          | 10.78                      | 0.293                 |
| 24      | 7          | 7.19                       | 0.005                 |
| 25+     | 16         | 11.16                      | 2.101                 |

Now add the last column:

$$\chi^2 = \sum \frac{(O-E)^2}{E} \approx 10.22$$

Bins = 18  $\Rightarrow$   $df = 17$ .

Then chi-square gives a p-value (here it would be about **0.89**, which is “fine”).

---

**Exactly what you do with YOUR data**

For your one window:

1. You have  $K_1, \dots, K_{500}$ .
2. Make bins exactly like above (0–8, 9, 10, ..., 24, 25+).
3. Compute **Observed**  $O$  by counting how many  $K_i$ 's fall in each bin.
4. Compute **Expected**  $E$  using Poisson(16) (formula above) for each bin (sum probabilities inside bin  $\times$  500).
5. Compute  $\chi^2 = \sum (O - E)^2 / E$ .
6.  $df = \#bins - 1$ .
7. Convert  $\chi^2$  to p-value.

If you paste your **500 K values** (or just the **frequency counts** like “K=0 occurs ..., K=1 occurs ...”), I’ll build the bin table for your actual run and compute  $\chi^2$ , df, and the p-value.

**You said:**

calculate E for each bin then tell >5 rule in easy way

**ChatGPT said:**

Thought for 27s

**How to calculate Expected  $E$  for each bin (Diehard Birthday Spacings, 1 window)**

You have:

- number of runs:  $n = 500$
- Poisson mean:  $\lambda = 16$

Poisson probability for exactly  $k$ :

$$P(K = k) = e^{-16} \frac{16^k}{k!}$$

**Expected count** for exactly  $k$  in 500 runs:

$$E(k) = 500 \times P(K = k)$$

**Expected count** for a bin (range) like  $a \leq K \leq b$ :

$$E(a-b) = 500 \times \sum_{k=a}^b P(K = k)$$

---

**Expected  $E$  values for Diehard-style bins**

A common Diehard binning for  $\lambda = 16, n = 500$  becomes:

- Bin1: 0–8
- then single bins: 9,10, ...,24
- last bin: 25 +

Here are the **Expected counts**  $E$  (for  $n = 500, \lambda = 16$ ):

| Bin (K) | Expected $E$ |
|---------|--------------|
| 0–8     | 10.9936      |
| 9       | 10.6555      |
| 10      | 17.0488      |
| 11      | 24.7983      |
| 12      | 33.0644      |
| 13      | 40.6947      |
| 14      | 46.5082      |
| 15      | 49.6088      |
| 16      | 49.6088      |
| 17      | 46.6906      |
| 18      | 41.5028      |
| 19      | 34.9497      |
| 20      | 27.9598      |
| 21      | 21.3027      |

| Bin (K) | Expected $E$ |
|---------|--------------|
| 22      | 15.4929      |
| 23      | 10.7776      |
| 24      | 7.1851       |
| 25+     | 11.1577      |

(These expected values add up to 500.)

---

### The “expected $\geq 5$ ” rule (super easy)

Chi-square needs each bin’s expected count not too small.

Rule: **every bin must have  $E \geq 5$ .**

### How bin 0–8 is created (example)

Look at expected counts for small  $k$ :

| k | Expected $E(k)$ |
|---|-----------------|
| 0 | 0.000056        |
| 1 | 0.000900        |
| 2 | 0.00720         |
| 3 | 0.0384          |
| 4 | 0.1536          |
| 5 | 0.4917          |
| 6 | 1.3111          |
| 7 | 2.9969          |



| k | Expected $E(k)$ |
|---|-----------------|
| 8 | 5.9937          |

Now add them from the left:

- up to 7:  $0.000056 + \dots + 2.9969 \approx 4.9999$  (still  $< 5$ )
- include 8: becomes  $\approx 10.9936$  ( $\geq 5$ )

✓ So Diehard merges **0–8** into one bin because smaller ones alone would be too tiny.

After that:

- $E(9) = 10.6555$  already  $\geq 5 \rightarrow$  bin “9” alone
- $E(10) = 17.0488$  already  $\geq 5 \rightarrow$  bin “10” alone  
...and so on, until the tail where it may merge again (here “25+”).

### How df is obtained

If total bins =  $B$ , then:

$$df = B - 1$$

Here bins = 18, so:

$$df = 17$$

If you give me your **Observed counts** in the same bins (0–8, 9, 10, ..., 24, 25+), I’ll compute the chi-square term-by-term and the final p-value with you.

**You said:**

this process i have to do 9 time for 9 window

**ChatGPT said:**

Yes — **exactly**.

You do the same process **9 times**, once for each 24-bit window.

**What happens for each window (repeat 9 times)**

For window  $s = 0, 1, \dots, 8$ :

1. Read **1024** 32-bit integers  $\rightarrow$  extract 24-bit birthdays using that window
2. Sort birthdays  $\rightarrow$  compute **1024 spacings** (including wrap-around)
3. Count duplicates in spacings  $\rightarrow$  that gives **one  $K$**

4. Repeat steps 1–3 **500 times** → get  $K_1, \dots, K_{500}$
5. Make bins, compute **Observed**  $O$  and **Expected**  $E$  (Poisson(16))
6. Compute chi-square → get **one p-value** for this window

So after 9 windows you have:

$$p_1, p_2, \dots, p_9$$

### Final step (original Diehard)

Run a **KS test** on these 9 p-values (they should look Uniform(0,1)).  
That gives one final p-value.

### Tiny checklist

- **Per window:** 500  $K$ 's → 1 chi-square p-value
- **All windows:** 9 p-values → 1 KS p-value

That's the whole Diehard Birthday Spacings test.

### You said:

how ks test works please explain after finding ps

### ChatGPT said:

After the 9 windows you have **9 p-values**:

$$p_1, p_2, \dots, p_9$$

### Why KS test?

If your RNG is good, each window's chi-square p-value should behave like a **Uniform(0,1)** random number.

So Diehard asks:

"Do these 9 numbers look like they came from Uniform(0,1)?"

That question is answered by the **Kolmogorov–Smirnov (KS) test**.

### KS test in the simplest steps

#### Step 1: Sort the 9 p-values

Example (toy):

$$[0.62, 0.10, 0.91, 0.44, 0.08, 0.73, 0.55, 0.30, 0.20]$$

Sorted:

$$p_{(1)} = 0.08, p_{(2)} = 0.10, p_{(3)} = 0.20, \dots, p_{(9)} = 0.91$$

### Step 2: Compare to “perfect uniform”

For Uniform(0,1), the ideal CDF is:

$$F(x) = x$$

For sorted data, the empirical CDF jumps like:

- at  $p_{(1)}$ : empirical CDF =  $1/9$
- at  $p_{(2)}$ : empirical CDF =  $2/9$
- ...
- at  $p_{(i)}$ : empirical CDF =  $i/9$

### Step 3: Compute the maximum gap $D$

At each  $i$ , compute two gaps:

1. gap just **after** the point:

$$D_i^+ = \frac{i}{9} - p_{(i)}$$

2. gap just **before** the point:

$$D_i^- = p_{(i)} - \frac{i-1}{9}$$

Then:

$$D = \max_i (\max(D_i^+, D_i^-))$$

So **D is the biggest mismatch** between your p-values and perfect uniform.

- If p-values are nicely spread  $\rightarrow D$  small.
- If p-values cluster near 0 or 1  $\rightarrow D$  large.

### Quick mini-example (with 5 values, to see it fast)

Say we had 5 p-values:

$$[0.02, 0.10, 0.35, 0.70, 0.95]$$

Sorted already.

For  $n = 5$ :

| $i$ | $p_{(i)}$ | $i/n$ | $(i/n - p_{(i)})$ | $(p_{(i)} - (i-1)/n)$               |
|-----|-----------|-------|-------------------|-------------------------------------|
| 1   | 0.02      | 0.20  | 0.18              | 0.02                                |
| 2   | 0.10      | 0.40  | 0.30              | $0.10 - 0.20 = -0.10 \rightarrow 0$ |
| 3   | 0.35      | 0.60  | 0.25              | $0.35 - 0.40 = -0.05 \rightarrow 0$ |
| 4   | 0.70      | 0.80  | 0.10              | $0.70 - 0.60 = 0.10$                |
| 5   | 0.95      | 1.00  | 0.05              | $0.95 - 0.80 = 0.15$                |

Max gap  $D = 0.30$ .

Then KS converts  $D$  into a final **KS p-value**:

- big  $D \rightarrow$  small KS p-value (bad)
- small  $D \rightarrow$  normal KS p-value (good)

---

**What does KS p-value mean here?**

- KS p-value very small (like  $< 0.01$ ): your 9 p-values are **not uniform**  $\rightarrow$  suspicious.
  - KS p-value not extreme: looks fine.
- 

**One-sentence summary**

**KS test sorts the 9 p-values and measures the largest deviation between their cumulative pattern and the ideal uniform cumulative pattern.**

If you paste your 9 p-values, I can compute  $D$  and the KS p-value step-by-step with you.

**You said:**

what is upper limit for  $d$

**ChatGPT said:**

For the KS statistic  $D$ :

**Absolute upper limit**

$$0 \leq D \leq 1$$

So the **maximum possible value is 1**.

**But with only  $n$  samples (here  $n = 9$ ), the *effective* maximum is:**

$$D_{\max} = 1 - \frac{1}{n}$$

So for Diehard's 9 p-values:

$$D_{\max} = 1 - \frac{1}{9} = \frac{8}{9} \approx 0.8889$$

### Why not 1 exactly for finite $n$ ?

Because the empirical CDF jumps in steps of size  $1/n$ . The biggest gap you can create between a step function and the line  $F(x) = x$  is just under 1.

So:

- Theoretical max: **1**
- For  $n = 9$ : practical max: **8/9  $\approx$  0.8889**

**You said:**

i have 1024x500x4x9 bytes then how each time(for 500 runs) 1024 32 bit integer coming each time is it random or sequence

**ChatGPT said:**

They come from your file **in sequence**.

### 1) Inside one window

For that window (fixed shift  $s$ ) Diehard does **500 runs** like this:

- **Run 1:** read the next **1024** 32-bit integers from the file  $\rightarrow$  make 1024 birthdays  $\rightarrow$  compute one  $K_1$
- **Run 2:** read the next **1024** integers (immediately after the previous ones)  $\rightarrow$  compute  $K_2$
- ...
- **Run 500:** read the next **1024** integers  $\rightarrow$  compute  $K_{500}$

So the file is consumed **sequentially** in blocks of 1024 integers.

☑ Nothing is "randomly selected" from the file.

The *randomness* is only in the **content** of the integers produced by your RNG when you generated the file.

### 2) What happens across the 9 windows

**Original Diehard behavior** (like the code you posted) is:

- Do window 1 (bits 1–24) using the file stream
- Then **close and reopen** the file (or rewind)

- Do window 2 (bits 2–25) using the **same integers again**, but extracting a different 24-bit slice
- Repeat until window 9

So in classic Diehard, the 9 windows usually reuse the **same 1024×500 integers**; only the **bit window** changes.

That’s why, in the original Diehard approach, the minimum file size is about:

$$1024 \times 500 \times 4 \approx 2,048,000 \text{ bytes } (\approx 2 \text{ MB})$$

(not 9× bigger), because it rewinds/reopens each window.

### 3) About your “1024 × 500 × 4 × 9” bytes

That size is needed **only if** you decide:

- “I want each window to use fresh, non-overlapping data (no rewind).”

Then yes, you’d need:

$$1024 \times 500 \times 4 \times 9 \approx 18.4 \text{ MB}$$

But **original Diehard doesn’t require that**.

### 4) So are the integers “random” or “sequence”?

- They are read in a **sequence** from the file.
- Whether they “look random” depends on how you **generated** the file.

If you tell me how you generated outputdat (binary bytes from RNG? or printed numbers as text?), I can tell you if Diehard is reading it correctly and whether you should worry about endianness / formatting.

**You said:**

now anything that i left

**ChatGPT said:**

You’ve got the full workflow now. The only things people usually “leave out” (and then the test gives confusing results) are these:

#### 1) Wrap-around spacing (must)

When you compute spacings from sorted birthdays, you must include the last gap:

$$S_m = (Y_{(1)} + N) - Y_{(m)}$$

Without this, the Birthday test is wrong. (Your original code likely missed it.)

## 2) File format must be truly binary

Diehard expects raw bytes.

If your outputdat was created with `fprintf(fp,"%u\n",x)` then it's **text**, not binary → the test will read garbage.

**Binary generation must look like:**

```
fwrite(&x, 4, 1, fp);
```

## 3) Endianness (byte order)

My reader assumes **little-endian** 32-bit words (most PCs).

If you generated file on a system with different byte order, you must swap bytes or change reader.

## 4) Enough data for 1 window

For classic Diehard (reusing file per window), minimum is:

$$1024 \times 500 \times 4 \approx 2 \text{ MB}$$

If file is smaller, you'll hit EOF.

## 5) Binning rule (expected ≥ 5)

You don't pick bins by hand; bins are formed so each bin's **expected** count ≥ 5.

Then:

$$df = \#bins - 1$$

## 6) Don't over-interpret one p-value

Even a good RNG sometimes gives p-values near 0 or 1 by chance.

You look for **repeated extremes** across many tests / many runs.

---

### Quick final checklist

- File is **binary** bytes, not text
- Read as 32-bit ints correctly (endianness ok)
- For each window: 500 blocks × 1024 ints sequentially
- Include **wrap-around spacing**
- Count duplicates →  $K$
- Chi-square GOF vs Poisson(16) → p-value
- Repeat 9 windows → 9 p-values
- KS test on those 9 p-values

If you tell me **how you created** outputdat (binary or text; and which RNG), I can tell you if anything else is risky.

**You said:**

When you compute spacings from sorted birthdays, you must include the last gap:  $S_m = (Y_{(1)} + N) - Y_{(m)}$  that is  $y_1 - y_n$

**ChatGPT said:**

In that formula, the symbols mean the **sorted birthdays**.

**You start with birthdays**

You have  $m = 1024$  birthdays:

$$Y_1, Y_2, \dots, Y_m$$

Each  $Y_i$  is a 24-bit number between 0 and  $N - 1$ , where  $N = 2^{24}$ .

**Then you sort them**

After sorting, we write them with parentheses:

$$Y_{(1)} \leq Y_{(2)} \leq \dots \leq Y_{(m)}$$

- $Y_{(1)}$  = the **smallest** birthday in the sample
- $Y_{(m)}$  = the **largest** birthday in the sample
- $Y_{(n)}$  (in general) = the **n-th smallest** birthday

So  $Y_{(1)}$  is not “Y1” from the file — it’s the **minimum after sorting**.

**Why the last gap uses them**

Normal gaps:

$$S_i = Y_{(i+1)} - Y_{(i)}, i = 1, \dots, m - 1$$

That gives  $m - 1$  gaps.

The missing part is the circular gap from the largest back to the smallest across the “end of the year”:

$$S_m = (Y_{(1)} + N) - Y_{(m)}$$

- Add  $N$  because after day  $N - 1$ , the next day is day 0 again (circular year).

**Tiny example**

Say  $N = 20$ , and birthdays are:

$$[2, 15, 7, 8]$$

Sorted:

$$Y_{(1)} = 2, Y_{(2)} = 7, Y_{(3)} = 8, Y_{(4)} = 15$$



Gaps:

- $S_1 = 7 - 2 = 5$
- $S_2 = 8 - 7 = 1$
- $S_3 = 15 - 8 = 7$
- wrap gap:

$$S_4 = (2 + 20) - 15 = 7$$

So total gaps = 4 (=m), and they sum to 20 (=N).

That's exactly what those symbols mean.

**You said:**

in diehard what is N

**ChatGPT said:**

In the Diehard **Birthday Spacings** test,  $N$  means:

the total number of possible “days in the year” (the range size for birthdays).

Diehard sets:

- no\_bits = 24
- so

$$N = 2^{24} = 16,777,216.$$

So each extracted “birthday” is a 24-bit integer:

$$Y \in \{0, 1, 2, \dots, N - 1\}.$$

That's why in the wrap-around spacing we use  $+N$ : it's like wrapping around a circular year of length  $N$ .

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